

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250279864

Kind Code

A1

Publication Date

September 04, 2025

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RESOURCE POOL CONFIGURATION FOR SIDELINK POSITIONING

Abstract

A computer-readable storage medium stores instructions to configure a UE for sidelink positioning communication in a 5G NR network, and to cause the UE to perform operations including decoding RRC signaling. LTE positioning protocol (EPP) signaling received from a base station for an in-coverage scenario, or sidelink positioning protocol (SEPP) signaling from a second UE, or pre-configuration for an out-of-coverage scenario. The EPP, RRC, or SEPP signaling or the pre-configuration respectively includes configuration information configuring a first resource pool associated with sidelink communications and a second resource pool associated with SL PRS. The first resource pool is multiplexed with the second resource pool. Sidelink data is encoded for a sidelink transmission to a second UE using a first time-frequency resource from the first resource pool. A first SL PRS is encoded for transmission to a second UE using a second time-frequency resource from the second resource pool.

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Family ID: 89618241

Appl. No.: 18/858896

Filed (or PCT Filed): July 03, 2023

PCT No.: PCT/US2023/026842

Related U.S. Application Data

Publication Classification

Int. Cl.: H04L5/00 (20060101); H04W64/00 (20090101); H04W72/40 (20230101)

U.S. Cl.:

CPC H04L5/0051 (20130101); H04W64/00 (20130101); H04W72/40 (20230101);

Background/Summary

PRIORITY CLAIM [0001] This application claims the benefit of priority to the following patent applications: [0002] U.S. Provisional Patent Application No. 63/390,391, filed Jul. 19, 2022, and entitled “RESOURCE POOL CONFIGURATION FOR SIDELINK POSITIONING;” and [0003] U.S. Provisional Patent Application No. 63/487,173, filed Feb. 27, 2023, and entitled “RESOURCE POOL CONFIGURATION FOR SIDELINK POSITIONING.” [0004] Each of the above-listed applications is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0005] Aspects pertain to wireless communications. Some aspects relate to wireless networks including 3GPP (Third Generation Partnership Project) networks, 3GPP LTE (Long Term Evolution) networks, 3GPP LTE-A (LTE Advanced) networks, (MultaFire, LTE-U), and fifth-generation (5G) networks including 5G new radio (NR) (or 5G-NR) networks, 5G-LTE networks such as 5G NR unlicensed spectrum (NR-U) networks and other unlicensed networks including Wi-Fi, CBRS (OnGo), etc. Other aspects are directed to techniques for configuring resource pools for sidelink positioning.

BACKGROUND

[0006] Mobile communications have evolved significantly from early voice systems to today's highly sophisticated integrated communication platform. With the increase in different types of devices communicating with various network devices, the usage of 3GPP LTE systems has increased. The penetration of mobile devices (user equipment or UEs) in modern society has continued to drive demand for a wide variety of networked devices in many disparate environments. Fifth-generation (5G) wireless systems are forthcoming and are expected to enable even greater speed, connectivity, and usability. Next-generation 5G networks (or NR networks) are expected to increase throughput, coverage, and robustness and reduce latency and operational and capital expenditures. 5G-NR networks will continue to evolve based on 3GPP LTE-Advanced with additional potential new radio access technologies (RATs) to enrich people's lives with seamless wireless connectivity solutions delivering fast, rich content and services. As the current cellular network frequency is saturated, higher frequencies, such as millimeter wave (mmWave) frequency, can be beneficial due to their high bandwidth.

[0007] Potential LTE operation in the unlicensed spectrum includes (and is not limited to) the LTE operation in the unlicensed spectrum via dual connectivity (DC), or DC-based LAA, and the standalone LTE system in the unlicensed spectrum, according to which LTE-based technology solely operates in the unlicensed spectrum without requiring an “anchor” in the licensed spectrum, called MultaFire. Further enhanced operation of LTE and NR systems in the licensed, as well as

unlicensed spectrum, is expected in future releases and 5G systems. Such enhanced operations can include techniques for configuring resource pools for sidelink (SL) positioning.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0008] In the figures, which are not necessarily drawn to scale, like numerals may describe similar components in different views. Like numerals having different letter suffixes may represent different instances of similar components. The figures illustrate generally, by way of example, but not by way of limitation, various aspects discussed in the present document.

[0009] FIG. 1A illustrates an architecture of a network, in accordance with some aspects.

[0010] FIG. 1B and FIG. 1C illustrate a non-roaming 5G system architecture in accordance with some aspects.

[0011] FIG. 2, FIG. 3, and FIG. 4 illustrate various systems, devices, and components that may implement aspects of disclosed embodiments.

[0012] FIG. 5 illustrates an example of sidelink positioning with anchor UEs and a target UE, in accordance with some aspects.

[0013] FIG. 6 illustrates multiplexing resource pools for SL communication and sidelink positioning reference signal (SL PRS) in time-division multiplexing (TDM) with different slots, in accordance with some aspects.

[0014] FIG. 7 illustrates multiplexing resource pools for SL communication and SL PRS in TDM in a slot, in accordance with some aspects.

[0015] FIG. 8 illustrates multiplexing resource pools for SL communication and SL PRS in frequency-division multiplexing (FDM), in accordance with some aspects.

[0016] FIG. 9 illustrates multiplexing multiple SL PRS dedicated resource pools in TDM and FDM manner, in accordance with some aspects.

[0017] FIG. 10 illustrates SL PRS resources with different bandwidths in an SL PRS dedicated resource pool, in accordance with some aspects.

[0018] FIG. 11 illustrates SL PRS resources in an SL PRS resource pool configured according to a first option, in accordance with some aspects.

[0019] FIG. 12 illustrates SL PRS resources in an SL PRS resource pool configured according to a second option, in accordance with some aspects.

[0020] FIG. 13 illustrates SL PRS resources in an SL PRS resource pool configured according to a third option, in accordance with some aspects.

[0021] FIG. 14 illustrates SL PRS resource determination within an SL PRS resource pool, in accordance with some aspects.

[0022] FIG. 15 illustrates a determination of a set of slots for the SL PRS resource pool according to a first option, in accordance with some aspects.

[0023] FIG. 16 illustrates a determination of a set of slots for the SL PRS resource pool according to a first option, in accordance with some aspects.

[0024] FIG. 17 illustrates a block diagram of a communication device such as an evolved Node-B (eNB), a new generation Node-B (gNB) (or another RAN node), an NCR, an access point (AP), a wireless station (STA), a mobile station (MS), or user equipment (UE), in accordance with some aspects.

DETAILED DESCRIPTION

[0025] The following description and the drawings sufficiently illustrate aspects to enable those skilled in the art to practice them. Other aspects may incorporate structural, logical, electrical, process, and other changes. Portions and features of some aspects may be included in or substituted for, those of other aspects. Aspects outlined in the claims encompass all available equivalents of

those claims.

[0026] FIG. 1A illustrates an architecture of a network in accordance with some aspects. The communication network **140A** is shown to include user equipment (UE) **101** and UE **102**. The UEs **101** and **102** are illustrated as smartphones (e.g., handheld touchscreen mobile computing devices connectable to one or more cellular networks) but may also include any mobile or non-mobile computing device, such as Personal Data Assistants (PDAs), pagers, laptop computers, desktop computers, wireless handsets, drones, or any other computing device including a wired and/or wireless communications interface. The UEs **101** and **102** can be collectively referred to herein as UE **101**, and UE **101** can be used to perform one or more of the techniques disclosed herein.

[0027] Any of the radio links described herein (e.g., as used in the network **140A** or any other illustrated network) may operate according to any exemplary radio communication technology and/or standard.

[0028] LTE and LTE-Advanced are standards for wireless communications of high-speed data for UE such as mobile telephones. In LTE-Advanced and various wireless systems, carrier aggregation is a technology according to which multiple carrier signals operating on different frequencies may be used to carry communications for a single UE, thus increasing the bandwidth available to a single device. In some aspects, carrier aggregation may be used where one or more component carriers operate on unlicensed frequencies.

[0029] Aspects described herein can be used in the context of any spectrum management scheme including, for example, dedicated licensed spectrum, unlicensed spectrum, (licensed) shared spectrum (such as Licensed Shared Access (LSA) in 2.3-2.4 GHz, 3.4-3.6 GHz, 3.6-3.8 GHz, and further frequencies and Spectrum Access System (SAS) in 3.55-3.7 GHz and further frequencies).

[0030] Aspects described herein can also be applied to different Single Carrier or OFDM flavors (CP-OFDM, SC-FDMA, SC-OFDM, filter bank-based multicarrier (FBMC), OFDMA, etc.) and in particular 3GPP NR (New Radio) by allocating the OFDM carrier data bit vectors to the corresponding symbol resources.

[0031] In some aspects, any of the UEs **101** and **102** can comprise an Internet-of-Things (IoT) UE or a Cellular IoT (CIoT) UE, which can comprise a network access layer designed for low-power IoT applications utilizing short-lived UE connections. In some aspects, any of the UEs **101** and **102** can include a narrowband (NB) IoT UE (e.g., such as an enhanced NB-IoT (eNB-IoT) UE and Further Enhanced (FeNB-IoT) UE). An IoT UE can utilize technologies such as machine-to-machine (M2M) or machine-type communications (MTC) for exchanging data with an MTC server or device via a public land mobile network (PLMN), Proximity-Based Service (ProSe), or device-to-device (D2D) communication, sensor networks, or IoT networks. The M2M or MTC exchange of data may be a machine-initiated exchange of data. An IoT network includes interconnecting IoT UEs, which may include uniquely identifiable embedded computing devices (within the Internet infrastructure), with short-lived connections. The IoT UEs may execute background applications (e.g., keep-alive messages, status updates, etc.) to facilitate the connections of the IoT network.

[0032] In some aspects, any of the UEs **101** and **102** can include enhanced MTC (eMTC) UEs or further enhanced MTC (FeMTC) UEs.

[0033] The UEs **101** and **102** may be configured to connect, e.g., communicatively coupled, with a radio access network (RAN) **110**. The RAN **110** may be, for example, a Universal Mobile Telecommunications System (UMTS), an Evolved Universal Terrestrial Radio Access Network (E-UTRAN), a NextGen RAN (NG RAN), or some other type of RAN. The UEs **101** and **102** utilize connections **103** and **104**, respectively, each of which comprises a physical communications interface or layer (discussed in further detail below); in this example, the connections **103** and **104** are illustrated as an air interface to enable communicative coupling and can be consistent with cellular communications protocols, such as a Global System for Mobile Communications (GSM) protocol, a code-division multiple access (CDMA) network protocol, a Push-to-Talk (PTT) protocol, a PTT over Cellular (POC) protocol, a Universal Mobile Telecommunications System

(UMTS) protocol, a 3GPP Long Term Evolution (LTE) protocol, a fifth-generation (5G) protocol, a New Radio (NR) protocol, and the like.

[0034] In an aspect, the UEs **101** and **102** may further directly exchange communication data via a ProSe interface **105**. The ProSe interface **105** may alternatively be referred to as a sidelink interface comprising one or more logical channels, including but not limited to a Physical Sidelink Control Channel (PSCCH), a Physical Sidelink Shared Channel (PSSCH), a Physical Sidelink Discovery Channel (PSDCH), and a Physical Sidelink Broadcast Channel (PSBCH).

[0035] The UE **102** is shown to be configured to access an access point (AP) **106** via connection **107**. The connection **107** can comprise a local wireless connection, such as, for example, a connection consistent with any IEEE 802.11 protocol, according to which the AP **106** can comprise a wireless fidelity (WiFi®) router. In this example, the AP **106** is shown to be connected to the Internet without connecting to the core network of the wireless system (described in further detail below).

[0036] The RAN **110** can include one or more access nodes that enable connections **103** and **104**. These access nodes (ANs) can be referred to as base stations (BSs), NodeBs, evolved NodeBs (eNBs), Next Generation NodeBs (gNBs), RAN network nodes, and the like, and can comprise ground stations (e.g., terrestrial access points) or satellite stations providing coverage within a geographic area (e.g., a cell). In some aspects, communication nodes **111** and **112** can be transmission/reception points (TRPs). In instances when the communication nodes **111** and **112** are NodeBs (e.g., eNBs or gNBs), one or more TRPs can function within the communication cell of the NodeBs. The RAN **110** may include one or more RAN nodes for providing macrocells, e.g., macro RAN node **111**, and one or more RAN nodes for providing femtocells or picocells (e.g., cells having smaller coverage areas, smaller user capacity, or higher bandwidth compared to macrocells), e.g., low power (LP) RAN node **112** or an unlicensed spectrum based secondary RAN node **112**.

[0037] Any of the RAN nodes **111** and **112** can terminate the air interface protocol and can be the first point of contact for the UEs **101** and **102**. In some aspects, any of the RAN nodes **111** and **112** can fulfill various logical functions for the RAN **110** including, but not limited to, the radio network controller (RNC) functions such as radio bearer management, uplink and downlink dynamic radio resource management, and data packet scheduling, and mobility management. In an example, any of the nodes **111** and/or **112** can be a new generation Node-B (gNB), an evolved node-B (eNB), or another type of RAN node.

[0038] The RAN **110** is shown to be communicatively coupled to a core network (CN) **120** via an S1 interface **113**. In aspects, the CN **120** may be an evolved packet core (EPC) network, a NextGen Packet Core (NPC) network, or some other type of CN (e.g., as illustrated in FIGS. **1B-1C**). In this aspect, the S1 interface **113** is split into two parts: the S1-U interface **114**, which carries user traffic data between the RAN nodes **111** and **112** and the serving gateway (S-GW) **122**, and the S1-mobility management entity (MME) interface **115**, which is a signaling interface between the RAN nodes **111** and **112** and MMEs **121**.

[0039] In this aspect, the CN **120** comprises the MMEs **121**, the S-GW **122**, the Packet Data Network (PDN) Gateway (P-GW) **123**, and a home subscriber server (HSS) **124**. The MMEs **121** may be similar in function to the control plane of legacy Serving General Packet Radio Service (GPRS) Support Nodes (SGSN). The MMEs **121** may manage mobility aspects in access such as gateway selection and tracking area list management. The HSS **124** may comprise a database for network users, including subscription-related information to support the network entities' handling of communication sessions. The CN **120** may comprise one or several HSSs **124**, depending on the number of mobile subscribers, the capacity of the equipment, the organization of the network, etc. For example, the HSS **124** can provide support for routing/roaming, authentication, authorization, naming/addressing resolution, location dependencies, etc.

[0040] The S-GW **122** may terminate the S1 interface **113** towards the RAN **110**, and route data

packets between the RAN **110** and the CN **120**. In addition, the S-GW **122** may be a local mobility anchor point for inter-RAN node handovers and also may provide an anchor for inter-3GPP mobility. Other responsibilities of the S-GW **122** may include lawful intercept, charging, and some policy enforcement.

[0041] The P-GW **123** may terminate an SGi interface toward a PDN. The P-GW **123** may route data packets between the EPC network **120** and external networks such as a network including the application server **184** (alternatively referred to as application function (AF)) via an Internet Protocol (IP) interface **125**. The P-GW **123** can also communicate data to other external networks **131A**, which can include the Internet, IP multimedia subsystem (IPS) network, and other networks. Generally, the application server **184** may be an element offering applications that use IP bearer resources with the core network (e.g., UMTS Packet Services (PS) domain, LTE PS data services, etc.). In this aspect, the P-GW **123** is shown to be communicatively coupled to an application server **184** via an IP interface **125**. The application server **184** can also be configured to support one or more communication services (e.g., Voice-over-Internet Protocol (VOIP) sessions, PTT sessions, group communication sessions, social networking services, etc.) for the UEs **101** and **102** via the CN **120**.

[0042] The P-GW **123** may further be a node for policy enforcement and charging data collection. Policy and Charging Rules Function (PCRF) **126** is the policy and charging control element of the CN **120**. In a non-roaming scenario, in some aspects, there may be a single PCRF in the Home Public Land Mobile Network (HPLMN) associated with a UE's Internet Protocol Connectivity Access Network (IP-CAN) session. In a roaming scenario with a local breakout of traffic, there may be two PCRFs associated with a UE's IP-CAN session: a Home PCRF (H-PCRF) within an HPLMN and a Visited PCRF (V-PCRF) within a Visited Public Land Mobile Network (VPLMN). The PCRF **126** may be communicatively coupled to the application server **184** via the P-GW **123**.

[0043] In some aspects, the communication network **140A** can be an IoT network or a 5G network, including a 5G new radio network using communications in the licensed (5G NR) and the unlicensed (5G NR-U) spectrum. One of the current enablers of IoT is the narrowband IoT (NB-IoT).

[0044] An NG system architecture can include the RAN **110** and a 5G core network **120**. RAN **110** in an NG system can be referred to as NG-RAN **110**. The NG-RAN **110** can include a plurality of nodes, such as gNBs and NG-eNBs. The core network **120** (also referred to as a 5G core network or 5GC) can include an access and mobility function (AMF) and/or a user plane function (UPF). The AMF and the UPF can be communicatively coupled to the gNBs and the NG-eNBs via NG interfaces. More specifically, in some aspects, the gNBs and the NG-eNBs can be connected to the AMF by NG-C interfaces, and the UPF by NG-U interfaces. The gNBs and the NG-eNBs can be coupled to each other via Xn interfaces.

[0045] In some aspects, the NG system architecture can use reference points between various nodes as provided by 3GPP Technical Specification (TS) 23.501 (e.g., V15.4.0, 2018-12). In some aspects, each of the gNBs and the NG-eNBs can be implemented as a base station, a mobile edge server, a small cell, a home eNB, a RAN network node, and so forth. In some aspects, a gNB can be a master node (MN) and NG-eNB can be a secondary node (SN) in a 5G architecture. In some aspects, the master/primary node may operate in a licensed band and the secondary node may operate in an unlicensed band.

[0046] FIG. **1B** illustrates a non-roaming 5G system architecture in accordance with some aspects. Referring to FIG. **1B**, there is illustrated a 5G system architecture **140B** in a reference point representation. More specifically, UE **102** can be in communication with RAN **110** as well as one or more other 5G core (5GC) network entities. The 5G system architecture **140B** includes a plurality of network functions (NFs), such as access and mobility management function (AMF) **132**, location management function (LMF) **133**, session management function (SMF) **136**, policy control function (PCF) **148**, application function (AF) **150**, user plane function (UPF) **134**, network

slice selection function (NSSF) **142**, authentication server function (AUSF) **144**, and unified data management (UDM)/home subscriber server (HSS) **146**. The UPF **134** can provide a connection to a data network (DN) **152**, which can include, for example, operator services, Internet access, or third-party services. The AMF **132** can be used to manage access control and mobility and can also include network slice selection functionality. The SMF **136** can be configured to set up and manage various sessions according to network policy. The UPF **134** can be deployed in one or more configurations according to the desired service type. The PCF **148** can be configured to provide a policy framework using network slicing, mobility management, and roaming (similar to PCRF in a 4G communication system). The UDM can be configured to store subscriber profiles and data (similar to an HSS in a 4G communication system).

[0047] The LMF **133** may be used in connection with 5G positioning functionalities. In some aspects, LMF **133** receives measurements and assistance information from the next-generation radio access network (NG-RAN) **110** and the mobile device (e.g., UE **101**) via the AMF **132** over the NLs interface to compute the position of the UE **101**. In some aspects, NR positioning protocol A (NRPPa) may be used to carry the positioning information between NG-RAN and LMF **133** over a next-generation control plane interface (NG-C). In some aspects, LMF **133** configures the UE using the LTE positioning protocol (LPP) via AMF **132**. The NG-RAN **110** configures the UE **101** using radio resource control (RRC) protocol over LTE-Uu and NR-Uu interfaces.

[0048] In some aspects, the 5G system architecture **140B** configures different reference signals to enable positioning measurements. Example reference signals that may be used for positioning measurements include the positioning reference signal (NR PRS) in the downlink and the sounding reference signal (SRS) for positioning in the uplink. The downlink positioning reference signal (PRS) is a reference signal configured to support downlink-based positioning methods.

[0049] In some aspects, the 5G system architecture **140B** includes an IP multimedia subsystem (IMS) **168B** as well as a plurality of IP multimedia core network subsystem entities, such as call session control functions (CSCFs). More specifically, the IMS **168B** includes a CSCF, which can act as a proxy CSCF (P-CSCF) **162B**, a serving CSCF (S-CSCF) **164B**, an emergency CSCF (E-CSCF) (not illustrated in FIG. 1B), or interrogating CSCF (I-CSCF) **166B**. The P-CSCF **162B** can be configured to be the first contact point for the UE **102** within the IMS **168B**. The S-CSCF **164B** can be configured to handle the session states in the network, and the E-CSCF can be configured to handle certain aspects of emergency sessions such as routing an emergency request to the correct emergency center or PSAP. The I-CSCF **166B** can be configured to function as the contact point within an operator's network for all IMS connections destined to a subscriber of that network operator, or a roaming subscriber currently located within that network operator's service area. In some aspects, the I-CSCF **166B** can be connected to another IP multimedia network **170**, e.g. an IMS operated by a different network operator.

[0050] In some aspects, the UDM/HSS **146** can be coupled to an application server (AS) **160B**, which can include a telephony application server (TAS) or another AS. The AS **160B** can be coupled to the IMS **168B** via the S-CSCF **164B** or the I-CSCF **166B**.

[0051] A reference point representation shows that interaction can exist between corresponding NF services. For example, FIG. 1B illustrates the following reference points: N1 (between the UE **102** and the AMF **132**), N2 (between the RAN **110** and the AMF **132**), N3 (between the RAN **110** and the UPF **134**), N4 (between the SMF **136** and the UPF **134**), N5 (between the PCF **148** and the AF **150**, not shown), N6 (between the UPF **134** and the DN **152**), N7 (between the SMF **136** and the PCF **148**, not shown), N8 (between the UDM **146** and the AMF **132**, not shown), N9 (between two UPFs **134**, not shown), N10 (between the UDM **146** and the SMF **136**, not shown), N11 (between the AMF **132** and the SMF **136**, not shown), N12 (between the AUSF **144** and the AMF **132**, not shown), N13 (between the AUSF **144** and the UDM **146**, not shown), N14 (between two AMFs **132**, not shown), N15 (between the PCF **148** and the AMF **132** in case of a non-roaming scenario, or between the PCF **148** and a visited network and AMF **132** in case of a roaming scenario, not

shown), N16 (between two SMFs, not shown), and N22 (between AMF **132** and NSSF **142**, not shown). Other reference point representations not shown in FIG. **1B** can also be used.

[0052] FIG. **1C** illustrates a 5G system architecture **140C** and a service-based representation. In addition to the network entities illustrated in FIG. **1B**, system architecture **140C** can also include a network exposure function (NEF) **154** and a network repository function (NRF) **156**. In some aspects, 5G system architectures can be service-based and interaction between network functions can be represented by corresponding point-to-point reference points Ni or as service-based interfaces.

[0053] In some aspects, as illustrated in FIG. **1C**, service-based representations can be used to represent network functions within the control plane that enable other authorized network functions to access their services. In this regard, 5G system architecture **140C** can include the following service-based interfaces: Namf **158H** (a service-based interface exhibited by the AMF **132**), Nsmf **158I** (a service-based interface exhibited by the SMF **136**), Nnef **158B** (a service-based interface exhibited by the NEF **154**), Npcf **158D** (a service-based interface exhibited by the PCF **148**), a Nudm **158E** (a service-based interface exhibited by the UDM **146**), Naf **158F** (a service-based interface exhibited by the AF **150**), Nnrf **158C** (a service-based interface exhibited by the NRF **156**), Nnssf **158A** (a service-based interface exhibited by the NSSF **142**), Nausf **158G** (a service-based interface exhibited by the AUSF **144**). Other service-based interfaces (e.g., Nudr, N5g-eir, and Nudsf) not shown in FIG. **1C** can also be used.

[0054] FIGS. **2-13** illustrate various systems, devices, and components that may implement aspects of disclosed embodiments in different communication systems, such as 5G-NR networks including 5G non-terrestrial networks (NTNs). UEs, base stations (such as gNBs), and/or other nodes (e.g., satellites or other NTN nodes) discussed herein can be configured to perform the disclosed techniques.

[0055] FIG. **2** illustrates a network **200** in accordance with various embodiments. The network **200** may operate in a manner consistent with 3GPP technical specifications for LTE or 5G/NR systems. However, the example embodiments are not limited in this regard and the described embodiments may apply to other networks that benefit from the principles described herein, such as future 3GPP systems, or the like.

[0056] The network **200** may include a UE **202**, which may include any mobile or non-mobile computing device designed to communicate with a RAN **204** via an over-the-air connection. The UE **202** may be but is not limited to, a smartphone, tablet computer, wearable computing device, desktop computer, laptop computer, in-vehicle infotainment, in-car entertainment device, instrument cluster, a head-up display device, onboard diagnostic device, dashtop mobile equipment, mobile data terminal, electronic engine management system, electronic/engine control unit, electronic/engine control module, embedded system, sensor, microcontroller, control module, engine management system, networked appliance, machine-type communication device, M2M or D2D device, IoT device, etc.

[0057] In some embodiments, network **200** may include a plurality of UEs coupled directly with one another via a sidelink interface. The UEs may be M2M/D2D devices that communicate using physical sidelink channels such as but not limited to, PSBCH, PSDCH, PSSCH, PSCCH, PSFCH, etc.

[0058] In some embodiments, the UE **202** may additionally communicate with an AP **206** via an over-the-air connection. The AP **206** may manage a WLAN connection, which may serve to offload some/all network traffic from the RAN **204**. The connection between the UE **202** and the AP **206** may be consistent with any IEEE 802.11 protocol, wherein the AP **206** could be a wireless fidelity (Wi-Fi®) router. In some embodiments, the UE **202**, RAN **204**, and AP **206** may utilize cellular-WLAN aggregation (for example, LWA/LWIP). Cellular-WLAN aggregation may involve the UE **202** configured by the RAN **204** to utilize both cellular radio resources and WLAN resources.

[0059] The RAN **204** may include one or more access nodes, for example, access node (AN) **208**.

AN **208** may terminate air-interface protocols for the UE **202** by providing access stratum protocols including RRC, Packet Data Convergence Protocol (PDCP), Radio Link Control (RLC), MAC, and L1 protocols. In this manner, the AN **208** may enable data/voice connectivity between the core network (CN) **220** and the UE **202**. In some embodiments, the AN **208** may be implemented in a discrete device or as one or more software entities running on server computers as part of, for example, a virtual network, which may be referred to as a CRAN or virtual baseband unit pool. The AN **208** be referred to as a BS, gNB, RAN node, eNB, ng-eNB, NodeB, RSU, TRxP, TRP, etc. The AN **208** may be a macrocell base station or a low-power base station for providing femtocells, picocells, or other like cells having smaller coverage areas, smaller user capacity, or higher bandwidth compared to macrocells.

[0060] In embodiments in which the RAN **204** includes a plurality of ANs, they may be coupled with one another via an X2 interface (if the RAN **204** is an LTE RAN) or an Xn interface (if the RAN **204** is a 5G RAN). The X2/Xn interfaces, which may be separated into control/user plane interfaces in some embodiments, may allow the ANs to communicate information related to handovers, data/context transfers, mobility, load management, interference coordination, etc.

[0061] The ANs of the RAN **204** may each manage one or more cells, cell groups, component carriers, etc. to provide the UE **202** with an air interface for network access. The UE **202** may be simultaneously connected with a plurality of cells provided by the same or different ANs of the RAN **204**. For example, the UE **202** and RAN **204** may use carrier aggregation to allow the UE **202** to connect with a plurality of component carriers, each corresponding to a Pcell or Scell. In dual connectivity scenarios, a first AN may be a master node that provides an MCG, and a second AN may be a secondary node that provides an SCG. The first/second ANs may be any combination of eNB, gNB, ng-eNB, etc.

[0062] The RAN **204** may provide the air interface over a licensed spectrum or an unlicensed spectrum. To operate in the unlicensed spectrum, the nodes may use LAA, eLAA, and/or feLAA mechanisms based on CA technology with PCells/SCells. Before accessing the unlicensed spectrum, the nodes may perform medium/carrier-sensing operations based on, for example, a listen-before-talk (LBT) protocol.

[0063] In V2X scenarios, the UE **202** or AN **208** may be or act as a roadside unit (RSU), which may refer to any transportation infrastructure entity used for V2X communications. An RSU may be implemented in or by a suitable AN or a stationary (or relatively stationary) UE. An RSU implemented in or by a UE may be referred to as a “UE-type RSU”; an eNB may be referred to as an “eNB-type RSU”; a gNB may be referred to as a “gNB-type RSU”; and the like. In one example, an RSU is a computing device coupled with radio frequency circuitry located on a roadside that provides connectivity support to passing vehicle UEs. The RSU may also include internal data storage circuitry to store intersection map geometry, traffic statistics, and media, as well as applications/software to sense and control ongoing vehicular and pedestrian traffic. The RSU may provide very low latency communications required for high-speed events, such as crash avoidance, traffic warnings, and the like. Additionally, or alternatively, the RSU may provide other cellular/WLAN communications services. The components of the RSU may be packaged in a weatherproof enclosure suitable for outdoor installation and may include a network interface controller to provide a wired connection (e.g., Ethernet) to a traffic signal controller or a backhaul network.

[0064] In some embodiments, the RAN **204** may be an LTE RAN **210** with eNBs, for example, eNB **212**. The LTE RAN **210** may provide an LTE air interface with the following characteristics: sub-carrier spacing (SCS) of 15 kHz; CP-OFDM waveform for downlink (DL) and SC-FDMA waveform for uplink (UL); turbo codes for data and TBCC for control; etc. The LTE air interface may rely on CSI-RS for CSI acquisition and beam management; PDSCH/PDCCH DMRS for PDSCH/PDCCH demodulation; and CRS for cell search and initial acquisition, channel quality measurements, and channel estimation for coherent demodulation/detection at the UE. The LTE air

interface may operate on sub-6 GHz bands.

[0065] In some embodiments, the RAN **204** may be an NG-RAN **214** with gNBs, for example, gNB **216**, or ng-eNBs, for example, ng-eNB **218**. The gNB **216** may connect with 5G-enabled UEs using a 5G NR interface. The gNB **216** may connect with a 5G core through an NG interface, which may include an N2 interface or an N3 interface. The ng-eNB **218** may also connect with the 5G core through an NG interface but may connect with a UE via an LTE air interface. The gNB **216** and the ng-eNB **218** may connect over an Xn interface.

[0066] In some embodiments, the NG interface may be split into two parts, an NG user plane (NG-U) interface, which carries traffic data between the nodes of the NG-RAN **214** and a UPF **248** (e.g., N3 interface), and an NG control plane (NG-C) interface, which is a signaling interface between the nodes of the NG-RAN **214** and an AMF **244** (e.g., N2 interface).

[0067] The NG-RAN **214** may provide a 5G-NR air interface with the following characteristics: variable SCS; CP-OFDM for DL, CP-OFDM, and DFT-s-OFDM for UL; polar, repetition, simplex, and Reed-Muller codes for control and LDPC for data. The 5G-NR air interface may rely on CSI-RS, PDSCH/PDCCH DMRS similar to the LTE air interface. The 5G-NR air interface may not use a CRS but may use PBCH DMRS for PBCH demodulation; PTRS for phase tracking for PDSCH and tracking reference signal for time tracking. The 5G-NR air interface may operate on FR1 bands that include sub-6 GHz bands or FR2 bands that include bands from 24.25 GHz to 52.6 GHz. The 5G-NR air interface may include a synchronization signal and physical broadcast channel (SS/PBCH) block (SSB) which is an area of a downlink resource grid that includes PSS/SSS/PBCH.

[0068] In some embodiments, the 5G-NR air interface may utilize BWPs (bandwidth parts) for various purposes. For example, BWP can be used for dynamic adaptation of the SCS. For example, the UE **202** can be configured with multiple BWPs where each BWP configuration has a different SCS. When a BWP change is indicated to the UE **202**, the SCS of the transmission is changed as well. Another use case example of BWP is related to power saving. In particular, multiple BWPs can be configured for the UE **202** with different amounts of frequency resources (for example, PRBs) to support data transmission under different traffic loading scenarios. A BWP containing a smaller number of PRBs can be used for data transmission with a small traffic load while allowing power saving at the UE **202** and in some cases at the gNB **216**. A BWP containing a larger number of PRBs can be used for scenarios with higher traffic loads.

[0069] The RAN **204** is communicatively coupled to CN **220** which includes network elements to provide various functions to support data and telecommunications services to customers/subscribers (for example, users of UE **202**). The components of the CN **220** may be implemented in one physical node or separate physical nodes. In some embodiments, NFV may be utilized to virtualize any or all of the functions provided by the network elements of the CN **220** onto physical compute/storage resources in servers, switches, etc. A logical instantiation of the CN **220** may be referred to as a network slice, and a logical instantiation of a portion of the CN **220** may be referred to as a network sub-slice.

[0070] In some embodiments, the CN **220** may be connected to the LTE radio network as part of the Enhanced Packet System (EPS) **222**, which may also be referred to as an EPC (or enhanced packet core). The EPC **222** may include MME **224**, SGW **226**, SGSN **228**, HSS **230**, PGW **232**, and PCRF **234** coupled with one another over interfaces (or “reference points”) as shown. Functions of the elements of the EPC **222** may be briefly introduced as follows.

[0071] The MME **224** may implement mobility management functions to track the current location of the UE **202** to facilitate paging, bearer activation/deactivation, handovers, gateway selection, authentication, etc.

[0072] The SGW **226** may terminate an S1 interface toward the RAN and route data packets between the RAN and the EPC **222**. The SGW **226** may be a local mobility anchor point for inter-RAN node handovers and also may provide an anchor for inter-3GPP mobility. Other

responsibilities may include lawful intercept, charging, and some policy enforcement.

[0073] The SGSN **228** may track the location of the UE **202** and perform security functions and access control. In addition, the SGSN **228** may perform inter-EPC node signaling for mobility between different RAT networks; PDN and S-GW selection as specified by MME **224**; MME selection for handovers; etc. The S3 reference point between the MME **224** and the SGSN **228** may enable user and bearer information exchange for inter-3GPP access network mobility in idle/active states.

[0074] The HSS **230** may include a database for network users, including subscription-related information to support the network entities' handling of communication sessions. The HSS **230** can provide support for routing/roaming, authentication, authorization, naming/addressing resolution, location dependencies, etc. An S6a reference point between the HSS **230** and the MME **224** may enable the transfer of subscription and authentication data for authenticating/authorizing user access to the LTE CN **220**.

[0075] The PGW **232** may terminate an SGi interface toward a data network (DN) **236** that may include an application/content server **238**. The PGW **232** may route data packets between the LTE CN **220** and the data network **236**. The PGW **232** may be coupled with the SGW **226** by an S5 reference point to facilitate user plane tunneling and tunnel management. The PGW **232** may further include a node for policy enforcement and charging data collection (for example, PCEF). Additionally, the SGi reference point between the PGW **232** and the data network **236** may be an operator external public, a private PDN, or an intra-operator packet data network, for example, for the provision of IMS services. The PGW **232** may be coupled with a PCRF **234** via a Gx reference point.

[0076] The PCRF **234** is the policy and charging control element of the LTE CN **220**. The PCRF **234** may be communicatively coupled to the app/content server **238** to determine appropriate QoS and charging parameters for service flows. The PCRF **234** may provision associated rules into a PCEF (via Gx reference point) with appropriate TFT and QCI.

[0077] In some embodiments, the CN **220** may be a 5GC **240**. The 5GC **240** may include an AUSF **242**, AMF **244**, SMF **246**, UPF **248**, NSSF **250**, NEF **252**, NRF **254**, PCF **256**, UDM **258**, and AF **260** coupled with one another over interfaces (or “reference points”) as shown. Functions of the elements of the 5GC **240** may be briefly introduced as follows.

[0078] The AUSF **242** may store data for the authentication of UE **202** and handle authentication-related functionality. The AUSF **242** may facilitate a common authentication framework for various access types. In addition to communicating with other elements of the 5GC **240** over reference points as shown, the AUSF **242** may exhibit a Nausf service-based interface.

[0079] The AMF **244** may allow other functions of the 5GC **240** to communicate with the UE **202** and the RAN **204** and to subscribe to notifications about mobility events with respect to the UE **202**. The AMF **244** may be responsible for registration management (for example, for registering UE **202**), connection management, reachability management, mobility management, lawful interception of AMF-related events, and access authentication and authorization. The AMF **244** may provide transport for SM messages between the UE **202** and the SMF **246**, and act as a transparent proxy for routing SM messages. AMF **244** may also provide transport for SMS messages between UE **202** and an SMSF. AMF **244** may interact with the AUSF **242** and the UE **202** to perform various security anchor and context management functions. Furthermore, AMF **244** may be a termination point of a RAN CP interface, which may include or be an N2 reference point between the RAN **204** and the AMF **244**; and the AMF **244** may be a termination point of NAS (N1) signaling, and perform NAS ciphering and integrity protection. AMF **244** may also support NAS signaling with the UE **202** over an N3 IWF interface.

[0080] The SMF **246** may be responsible for SM (for example, session establishment, tunnel management between UPF **248** and AN **208**); UE IP address allocation and management (including optional authorization); selection and control of UP function; configuring traffic steering at UPF

248 to route traffic to proper destination; termination of interfaces toward policy control functions; controlling part of policy enforcement, charging, and QoS; lawful intercept (for SM events and interface to LI system); termination of SM parts of NAS messages; downlink data notification; initiating AN specific SM information, sent via AMF **244** over N2 to AN **208**; and determining SSC mode of a session. SM may refer to the management of a PDU session, and a PDU session or “session” may refer to a PDU connectivity service that provides or enables the exchange of PDUs between the UE **202** and the data network **236**.

[0081] The UPF **248** may act as an anchor point for intra-RAT and inter-RAT mobility, an external PDU session point of interconnecting to data network **236**, and a branching point to support multi-homed PDU sessions. The UPF **248** may also perform packet routing and forwarding, perform packet inspection, enforce the user plane part of policy rules, lawfully intercept packets (UP collection), perform traffic usage reporting, perform QoS handling for a user plane (e.g., packet filtering, gating, UL/DL rate enforcement), perform uplink traffic verification (e.g., SDF-to-QoS flow mapping), transport level packet marking in the uplink and downlink, and perform downlink packet buffering and downlink data notification triggering. UPF **248** may include an uplink classifier to support routing traffic flows to a data network.

[0082] The NSSF **250** may select a set of network slice instances serving the UE **202**. The NSSF **250** may also determine allowed NSSAI and the mapping to the subscribed S-NSSAIs if needed. The NSSF **250** may also determine the AMF set to be used to serve the UE **202**, or a list of candidate AMFs based on a suitable configuration and possibly by querying the NRF **254**. The selection of a set of network slice instances for the UE **202** may be triggered by the AMF **244** with which the UE **202** is registered by interacting with the NSSF **250**, which may lead to a change of AMF. The NSSF **250** may interact with the AMF **244** via an N22 reference point and may communicate with another NSSF in a visited network via an N31 reference point (not shown). Additionally, the NSSF **250** may exhibit an Nnssf service-based interface.

[0083] The NEF **252** may securely expose services and capabilities provided by 3GPP network functions for the third party, internal exposure/re-exposure, AFs (e.g., AF **260**), edge computing or fog computing systems, etc. In such embodiments, the NEF **252** may authenticate, authorize, or throttle the AFs. NEF **252** may also translate information exchanged with the AF **260** and information exchanged with internal network functions. For example, the NEF **252** may translate between an AF-Service-Identifier and an internal 5GC information. NEF **252** may also receive information from other NFs based on the exposed capabilities of other NFs. This information may be stored at the NEF **252** as structured data, or a data storage NF using standardized interfaces. The stored information can then be re-exposed by the NEF **252** to other NFs and AFs, or used for other purposes such as analytics. Additionally, the NEF **252** may exhibit a Nnef service-based interface.

[0084] The NRF **254** may support service discovery functions, receive NF discovery requests from NF instances, and provide information on the discovered NF instances to the NF instances. NRF **254** also maintains information on available NF instances and their supported services. As used herein, the terms “instantiate,” “instantiation,” and the like may refer to the creation of an instance, and an “instance” may refer to a concrete occurrence of an object, which may occur, for example, during the execution of program code. Additionally, the NRF **254** may exhibit the Nnrf service-based interface.

[0085] The PCF **256** may provide policy rules to control plane functions to enforce them, and may also support a unified policy framework to govern network behavior. The PCF **256** may also implement a front end to access subscription information relevant to policy decisions in a UDR of the UDM **258**. In addition to communicating with functions over reference points as shown, the PCF **256** exhibits an Npcf service-based interface.

[0086] The UDM **258** may handle subscription-related information to support the network entities' handling of communication sessions and may store the subscription data of UE **202**. For example, subscription data may be communicated via an N8 reference point between the UDM **258** and the

AMF **244**. The UDM **258** may include two parts, an application front end, and a UDR. The UDR may store subscription data and policy data for the UDM **258** and the PCF **256**, and/or structured data for exposure and application data (including PFDs for application detection, and application request information for multiple UEs **202**) for the NEF **252**. The Nudr service-based interface may be exhibited by the UDR to allow the UDM **258**, PCF **256**, and NEF **252** to access a particular set of the stored data, as well as to read, update (e.g., add, modify), delete, and subscribe to the notification of relevant data changes in the UDR. The UDM may include a UDM-FE, which is in charge of processing credentials, location management, subscription management, and so on. Several different front ends may serve the same user in different transactions. The UDM-FE accesses subscription information stored in the UDR and performs authentication credential processing, user identification handling, access authorization, registration/mobility management, and subscription management. In addition to communicating with other NFs over reference points as shown, the UDM **258** may exhibit the Nudm service-based interface.

[0087] The AF **260** may provide application influence on traffic routing, provide access to NEF, and interact with the policy framework for policy control.

[0088] In some embodiments, the 5GC **240** may enable edge computing by selecting operator/3rd party services to be geographically close to a point that the UE **202** is attached to the network. This may reduce latency and load on the network. To provide edge-computing implementations, the 5GC **240** may select a UPF **248** close to the UE **202** and execute traffic steering from the UPF **248** to data network **236** via the N6 interface. This may be based on the UE subscription data, UE location, and information provided by the AF **260**. In this way, the AF **260** may influence UPF (re) selection and traffic routing. Based on operator deployment, when AF **260** is considered to be a trusted entity, the network operator may permit AF **260** to interact directly with relevant NFs. Additionally, the AF **260** may exhibit a Naf service-based interface.

[0089] The data network **236** may represent various network operator services, Internet access, or third-party services that may be provided by one or more servers including, for example, application/content server **238**.

[0090] In some aspects, network **200** is configured for NR positioning using the location management function (LMF) **245**, which can be configured as an LMF node or as functionality in a different type of node. In some embodiments, LMF **245** is configured to receive measurements and assistance information from NG-RAN **214** and UE **202** via the AMF **244** (e.g., using an NLs interface) to compute the position of the UE. In some embodiments, NR positioning protocol A (NRPPa) protocol can be used for carrying the positioning information between NG-RAN **214** and LMF **245** over a next-generation control plane interface (NG-C). In some embodiments, LMF **245** configures the UE **202** using LTE positioning protocol (LPP) (e.g., LPP-based communication link) via the AMF **244**. In some aspects, NG-RAN **214** configures the UE **202** using, e.g., radio resource control (RRC) protocol signaling over, e.g., LTE-Uu and NR-Uu interfaces. In some aspects, UE **202** uses the LTE-Uu interface to communicate with the ng-eNB **218** and the NR-Uu interface to communicate with the gNB **216**. In some aspects, ng-eNB **216** and gNB **216** use NG-C interfaces to communicate with the AMF **244**.

[0091] In some embodiments, the following reference signals can be used to achieve positioning measurements in NR communication networks: NR positioning reference signal (NR PRS) in the downlink and sounding reference signal (SRS) for positioning in the uplink. The downlink positioning reference signal (PRS) can be used as a reference signal supporting downlink-based positioning techniques. In some aspects, the entire NR bandwidth can be covered by transmitting PRS over multiple symbols that can be aggregated to accumulate power.

[0092] FIG. **3** schematically illustrates a wireless network **300** in accordance with various embodiments. The wireless network **300** may include a UE **302** in wireless communication with AN **304**. The UE **302** and AN **304** may be similar to, and substantially interchangeable with, like-named components described elsewhere herein.

[0093] The UE **302** may be communicatively coupled with the AN **304** via connection **306**. Connection **306** is illustrated as an air interface to enable communicative coupling and can be consistent with cellular communications protocols such as an LTE protocol or a 5G NR protocol operating at mmWave or sub-6 GHz frequencies.

[0094] The UE **302** may include a host platform **308** coupled with a modem platform **310**. The host platform **308** may include application processing circuitry **312**, which may be coupled with protocol processing circuitry **314** of the modem platform **310**. The application processing circuitry **312** may run various applications for the UE **302** that source/sink application data. The application processing circuitry **312** may further implement one or more layer operations to transmit/receive application data to/from a data network. These layer operations may include transport (for example UDP) and Internet (for example IP) operations.

[0095] The protocol processing circuitry **314** may implement one or more layer operations to facilitate the transmission or reception of data over connection **306**. The layer operations implemented by the protocol processing circuitry **314** may include, for example, MAC, RLC, PDCP, RRC, and NAS operations.

[0096] The modem platform **310** may further include digital baseband circuitry **316** that may implement one or more layer operations that are “below” layer operations performed by the protocol processing circuitry **314** in a network protocol stack. These operations may include, for example, PHY operations including one or more of HARQ-ACK functions, scrambling/descrambling, encoding/decoding, layer mapping/de-mapping, modulation symbol mapping, received symbol/bit metric determination, multi-antenna port precoding/decoding, which may include one or more of space-time, space-frequency or spatial coding, reference signal generation/detection, preamble sequence generation and/or decoding, synchronization sequence generation/detection, control channel signal blind decoding, and other related functions.

[0097] The modem platform **310** may further include transmit circuitry **318**, receive circuitry **320**, RF circuitry **322**, and RF front end (RFFE) **324**, which may include or connect to one or more antenna panels **326**. Briefly, the transmit circuitry **318** may include a digital-to-analog converter, mixer, intermediate frequency (IF) components, etc.; the receive circuitry **320** may include an analog-to-digital converter, mixer, IF components, etc.; the RF circuitry **322** may include a low-noise amplifier, a power amplifier, power tracking components, etc.; RFFE **324** may include filters (for example, surface/bulk acoustic wave filters), switches, antenna tuners, beamforming components (for example, phase-array antenna components), etc. The selection and arrangement of the components of the transmit circuitry **318** receive circuitry **320**, RF circuitry **322**, RFFE **324**, and antenna panels **326** (referred to generically as “transmit/receive components”) may be specific to details of a specific implementation such as, for example, whether the communication is TDM or FDM, in mmWave or sub-6 GHz frequencies, etc. In some embodiments, the transmit/receive components may be arranged in multiple parallel transmit/receive chains, may be disposed of in the same or different chips/modules, etc.

[0098] In some embodiments, the protocol processing circuitry **314** may include one or more instances of control circuitry (not shown) to provide control functions for the transmit/receive components.

[0099] A UE reception may be established by and via the antenna panels **326**, RFFE **324**, RF circuitry **322**, receive circuitry **320**, digital baseband circuitry **316**, and protocol processing circuitry **314**. In some embodiments, the antenna panels **326** may receive a transmission from the AN **304** by receive-beamforming signals received by a plurality of antennas/antenna elements of the one or more antenna panels **326**.

[0100] A UE transmission may be established by and via the protocol processing circuitry **314**, digital baseband circuitry **316**, transmit circuitry **318**, RF circuitry **322**, RFFE **324**, and antenna panels **326**. In some embodiments, the transmit components of the UE **302** may apply a spatial filter to the data to be transmitted to form a transmit beam emitted by the antenna elements of the

antenna panels **326**.

[0101] Similar to the UE **302**, the AN **304** may include a host platform **328** coupled with a modem platform **330**. The host platform **328** may include application processing circuitry **332** coupled with protocol processing circuitry **334** of the modem platform **330**. The modem platform may further include digital baseband circuitry **336**, transmit circuitry **338**, receive circuitry **340**, RF circuitry **342**, RFFE circuitry **344**, and antenna panels **346**. The components of the AN **304** may be similar to and substantially interchangeable with the like-named components of the UE **302**. In addition to performing data transmission/reception as described above, the components of the AN **304** may perform various logical functions that include, for example, RNC functions such as radio bearer management, uplink and downlink dynamic radio resource management, and data packet scheduling.

[0102] FIG. **4** is a block diagram illustrating components, according to some example embodiments, able to read instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium) and perform any one or more of the methodologies discussed herein. Specifically, FIG. **4** shows a diagrammatic representation of hardware resources **400** including one or more processors (or processor cores) **410**, one or more memory/storage devices **420**, and one or more communication resources **430**, each of which may be communicatively coupled via a bus **440** or other interface circuitry. For embodiments where node virtualization (e.g., NFV) is utilized, a hypervisor **402** may be executed to provide an execution environment for one or more network slices/sub-slices to utilize the hardware resources **400**.

[0103] The one or more processors **410** may include, for example, a processor **412** and a processor **414**. The one or more processors **410** may be, for example, a central processing unit (CPU), a reduced instruction set computing (RISC) processor, a complex instruction set computing (CISC) processor, a graphics processing unit (GPU), a DSP such as a baseband processor, an ASIC, an FPGA, a radio-frequency integrated circuit (RFIC), another processor (including those discussed herein), or any suitable combination thereof.

[0104] The memory/storage devices **420** may include a main memory, disk storage, or any suitable combination thereof. The memory/storage devices **420** may include but are not limited to, any type of volatile, non-volatile, or semi-volatile memory such as dynamic random access memory (DRAM), static random access memory (SRAM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), Flash memory, solid-state storage, etc.

[0105] The communication resources **430** may include interconnection or network interface controllers, components, or other suitable devices to communicate with one or more peripheral devices **404** or one or more databases **406** or other network elements via a network **408**. For example, the communication resources **430** may include wired communication components (e.g., for coupling via USB, Ethernet, etc.), cellular communication components, NFC components, Bluetooth® (or Bluetooth® Low Energy) components, Wi-Fi® components, and other communication components.

[0106] Instructions **450** may comprise software, a program, an application, an applet, an app, or other executable code for causing at least any of the one or more processors **410** to perform any one or more of the methodologies discussed herein. The instructions **450** may reside, completely or partially, within at least one of the one or more processors **410** (e.g., within the processor's cache memory), the memory/storage devices **420**, or any suitable combination thereof. Furthermore, any portion of the instructions **450** may be transferred to the hardware resources **400** from any combination of the peripheral devices **404** or the databases **406**. Accordingly, the memory of the one or more processors **410**, the memory/storage devices **420**, the peripheral devices **404**, and the databases **406** are examples of computer-readable and machine-readable media.

[0107] For one or more embodiments, at least one of the components outlined in one or more of the

figures may be configured to perform one or more operations, techniques, processes, and/or methods as outlined in the example sections below. For example, baseband circuitry associated with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below. For another example, circuitry associated with a UE, base station, satellite, network element, etc. as described in connection with one or more of the figures may be configured to operate in accordance with the disclosed techniques including one or more of the examples set forth below in the example section.

[0108] The term “application” may refer to a complete and deployable package, or environment to achieve a certain function in an operational environment. The term “AI/ML application” or the like may be an application that contains some artificial intelligence (AI)/machine learning (ML) models and application-level descriptions. In some embodiments, an AI/ML application may be used for configuring or implementing one or more of the disclosed aspects.

[0109] The term “machine learning” or “ML” refers to the use of computer systems implementing algorithms and/or statistical models to perform a specific task(s) without using explicit instructions but instead relying on patterns and inferences. ML algorithms build or estimate mathematical model(s) (referred to as “ML models” or the like) based on sample data (referred to as “training data,” “model training information,” or the like) to make predictions or decisions without being explicitly programmed to perform such tasks. Generally, an ML algorithm is a computer program that learns from experience concerning some task and some performance measure, and an ML model may be any object or data structure created after an ML algorithm is trained with one or more training datasets. After training, an ML model may be used to make predictions on new datasets. Although the term “ML algorithm” refers to different concepts than the term “ML model,” these terms as discussed herein may be used interchangeably for the present disclosure.

[0110] The term “machine learning model,” “ML model,” or the like may also refer to ML methods and concepts used by an ML-assisted solution. An “ML-assisted solution” is a solution that addresses a specific use case using ML algorithms during operation. ML models include supervised learning (e.g., linear regression, k-nearest neighbor (KNN), decision tree algorithms, support machine vectors, Bayesian algorithm, ensemble algorithms, etc.) unsupervised learning (e.g., K-means clustering, principal component analysis (PCA), etc.), reinforcement learning (e.g., Q-learning, multi-armed bandit learning, deep RL, etc.), neural networks, and the like. Depending on the implementation a specific ML model could have many sub-models as components and the ML model may train all sub-models together. Separately trained ML models can also be chained together in an ML pipeline during inference. An “ML pipeline” is a set of functionalities, functions, or functional entities specific to an ML-assisted solution; an ML pipeline may include one or several data sources in a data pipeline, a model training pipeline, a model evaluation pipeline, and an actor. The “actor” is an entity that hosts an ML-assisted solution using the output of the ML model inference). The term “ML training host” refers to an entity, such as a network function, that hosts the training of the model. The term “ML inference host” refers to an entity, such as a network function, that hosts the model during inference mode (which includes both the model execution as well as any online learning if applicable). The ML host informs the actor about the output of the ML algorithm, and the actor decides on an action (an “action” is performed by an actor as a result of the output of an ML-assisted solution). The term “model inference information” refers to information used as an input to the ML model for determining inference(s); the data used to train an ML model and the data used to determine inferences may overlap, however, “training data” and “inference data” refer to different concepts.

[0111] Mobile communication has evolved significantly from early voice systems to today's highly sophisticated integrated communication platform. The next generation wireless communication systems, (e.g., 5G NR systems) will provide access to information and sharing of data anywhere, anytime by various users and applications. 5G NR is expected to be a unified network/system that targets to meet vastly different and sometimes conflicting performance dimensions and services.

Such diverse multi-dimensional requirements are driven by different services and applications. In general, 5G NR will evolve based on 3GPP LTE-Advanced with additional potential new Radio Access Technologies (RATs) to enrich people's lives with better, simple, and seamless wireless connectivity solutions. NR will enable everything connected by wireless and deliver fast, rich content and services.

[0112] 5G NR supports highly precise positioning in the vertical and horizontal dimensions, which relies on timing-based, angle-based, power-based, or hybrid techniques to estimate the user location in the network. In particular, the following RAT-dependent positioning techniques can be used to meet the positioning requirements for various use cases, e.g., indoor, outdoor, and Industrial Internet-of-Thing (IoT): [0113] (a) Downlink time difference of arrival (DL-TDOA); [0114] (b) Uplink time difference of arrival (UL-TDOA); [0115] (c) Downlink angle of departure (DL-AoD); [0116] (d) Uplink angle of arrival (UL AoA); [0117] (e) Multi-cell round trip time (multi-RTT); and [0118] (f) NR enhanced cell ID (E-CID).

[0119] With wide bandwidth for positioning signal and beamforming capability in the mmWave frequency band, higher positioning accuracy can be achieved by RAT-dependent positioning techniques. In Rel-16, downlink positioning reference signal (DL-PRS) and uplink sounding reference signal (UL-SRS) for positioning were introduced as enablers to achieve target performance characteristics.

[0120] In Rel-18, to address use cases such as autonomous driving, sidelink, or vehicle-to-everything (V2X) based positioning is considered. More specifically, various scenarios including in-coverage, partial coverage, and out-of-network coverage can be considered for sidelink positioning. In some aspects, to meet the positioning accuracy requirement, a new sidelink reference signal, i.e., sidelink position reference signal (SL PRS), can be introduced.

[0121] FIG. 5 illustrates diagram 500 of an example of sidelink positioning with anchor UEs and a target UE, in accordance with some aspects. In the example of FIG. 5, a target UE indicates the UE to be positioned while anchor

[0122] UEs indicate the UEs supporting positioning of the target UE (e.g., by transmitting and/or receiving SL PRS and providing positioning-related information such as measurements based on the SL PRS). In some aspects, the SL PRS can be transmitted between anchor and target UEs for sidelink positioning.

[0123] In Rel-16, the resource pool can be defined for sidelink communication, where multiple UEs share the same resource pool for sidelink channel/signal transmission. For mode 2 resource allocation, the UEs autonomously select resources within the resource pool for physical sidelink control channel (PSCCH) and physical sidelink shared channel (PSSCH) transmission. In mode 1 resource allocation, the base station controls the resource allocation to the UEs. For sidelink positioning, a dedicated resource pool can be configured for SL PRS transmission. In this case, certain mechanisms can be defined on the configuration of the dedicated resource pool for SL PRS transmission.

[0124] The disclosed techniques include mechanisms for the configuration of the resource pool for sidelink positioning. In particular, the disclosed techniques include resource pool configuration for SL PRS, SL PRS resource indexing within a resource pool, and the determination of a set of slots for an SL PRS resource pool.

Resource Pool Configuration for SL PRS

[0125] As mentioned above, to address use cases such as autonomous driving, sidelink, or vehicle-to-everything (V2X) based positioning is considered. More specifically, various scenarios including in-coverage, partial coverage, and out-of-network coverage can be considered for sidelink positioning. To meet the positioning accuracy requirement, a new sidelink reference signal, i.e., sidelink position reference signal (SL PRS) can be introduced.

[0126] In an embodiment, a dedicated resource pool for SL PRS may include both SL PRS as well as PSCCH associated with SL PRS transmissions without any associated PSSCH. In another

embodiment, a dedicated resource pool for SL PRS may only include SL PRS resources. In the following embodiments, the SL PRS resource pool may refer to a dedicated resource pool for SL PRS.

[0127] Embodiments of resource pool configurations for SL PRS are provided as follows.

[0128] In some aspects, multiple resource pools can be configured for SL PRS transmission or reception. Further, different transmit (Tx) and receive (Rx) resource pools can be configured for SL PRS transmission and reception, respectively. In particular, multiple resource pools may be multiplexed in a time-division multiplexing (TDM) and/or frequency-division multiplexing (FDM) manner.

[0129] In some aspects, for an SL PRS resource pool, a contiguous number of physical resource blocks (PRB) can be (pre-)configured in the frequency domain. Further, the starting PRB of an SL PRS resource pool can be (pre-) configured relative to point-A. In another option, the starting PRB of an SL PRS resource pool can be (pre-)configured relative to the starting PRB of the SL bandwidth part (BWP) for SL communication or channel bandwidth.

[0130] Alternatively, or in addition to indicating a number of PRBs, the frequency domain allocation of an SL PRS resource pool may be provided to a UE via (pre-)configuration of a number of subchannels where a subchannel may be further configured with size from one of {10, 12, 15, 20, 25, 50, 75, 100} contiguous-in-frequency PRBs.

[0131] In some aspects, the SL PRS resource pool may be located within or outside the SL-BWP for SL communication, or partially overlapping with the SL-BWP for SL communication. As a further extension, when the starting PRB and number of PRBs are not configured for the SL PRS resource pool, a default frequency domain resource configuration can be equal to the SL BWP for SL communication or dedicated SL BWP for SL PRS transmission.

[0132] In some embodiments, the starting PRB of an SL PRS resource pool can be (pre-)configured relative to the starting PRB of an SL BWP which is (pre-) configured for SL PRS transmission and reception. In some aspects, the SL-BWPs for SL communication and SL PRS transmission can be different.

[0133] Based on the Rel. 16 definition of SL BWP as well as SL resource pools several configurations can be changed to allow for support of separate SL-BWP for SL positioning. For backward compatibility, the SL BWP that is configured may only be used for SL communication that uses all available symbols of a slot. The corresponding resource pool would then be configured to only use the slots that do not overlap with the slots partially used for positioning.

[0134] To allow for slots that are partially used for positioning either two SL BWP with orthogonal symbol allocation or a new SL BWP for partial usage as positioning can be defined. Rel. 16/17 SL devices only have one active SL BWP; thus, they may only use the resources defined for communication only. Configuring multiple SL BWPs with different time configurations and activating them at the UE simultaneously is possible from the RF front-end perspective but is currently not standardized in this fashion. Alternatively, a UE may only be expected to maintain a single SL-BWP-either for SL communication or for SL positioning at a time, with a BWP switching mechanism used to switch between the two. Such switching may be realized based on the time domain configuration of the SL communication and SL positioning resource pools within the respective SL-BWPs provided to a UE via (pre-)configuration or based on dynamic indication via DCI from the serving cell or via SCI from another UE, or a combination of the two.

[0135] Further, a starting symbol and length of symbols in a slot can be (pre-)configured in the time domain for an SL PRS resource pool.

[0136] In addition, subcarrier spacing and/or cyclic prefixes can be (pre-)configured for an SL PRS resource pool. In case when starting PRB of the SL PRS resource pool is defined relative to the starting PRB of the SL BWP for SL communication, subcarrier spacing and/or cyclic prefix may be the same as that (pre-)configured for SL BWP for SL communication. In another option, subcarrier spacing and/or cyclic prefix can be determined in accordance with the configuration from SL BWP

for SL PRS transmission and reception

[0137] In some embodiments, an SL PRS resource pool may be multiplexed with an SL communication resource pool in a TDM and/or FDM manner or both. In some aspects, the number of PRBs and the number of symbols allocated for the resource pools for SL communication and SL PRS transmission may be different.

[0138] In aspects when an SL PRS resource pool is multiplexed with an SL communication resource pool in a TDM manner, in one option, an SL PRS resource pool may be (pre-)configured in a different slot from the slots that are (pre-)configured for an SL communication resource pool. In another option, an SL PRS resource pool may be (pre-)configured in different symbols in the same slot as the slots that are (pre-)configured for an SL communication resource pool.

[0139] FIG. 6 illustrates a diagram **600** of multiplexing resource pools for SL communication and sidelink positioning reference signal (SL PRS) in time-division multiplexing (TDM) with different slots, in accordance with some aspects. In the example of FIG. 6, the SL PRS resource pool can be configured in a disjoint set of slots from the resource pool for SL communication. In some aspects, this is possible to be configured by orthogonal resource pools. In addition, there is the potential that the SL PRS is designed in a way that they could occupy a larger portion of the guard band, while still satisfying the spectral mask requirements.

[0140] FIG. 7 illustrates a diagram **700** of multiplexing resource pools for SL communication and SL PRS in TDM in a slot, in accordance with some aspects. In the example of FIG. 7, the SL PRS resource pool can be configured in the same slot but with a disjoint set of symbols from the resource pool for SL communication. This may be possible when the SL PRS resource pool and SL communication resource pool are configured in different SL BWPs.

[0141] FIG. 8 illustrates a diagram **800** of multiplexing resource pools for SL communication and SL PRS in frequency-division multiplexing (FDM), in accordance with some aspects. In the example of FIG. 8, the SL PRS resource pool can be configured in the same slot as the resource pool for SL communication. Further, the SL PRS resource pool can be multiplexed with the SL communication resource pool in an FDM manner. In some aspects, although, as shown in FIG. 8, the same set of symbols is allocated for the resource pools for SL communication and SL PRS, a different set of symbols can be straightforwardly extended from the example.

[0142] In some embodiments, one or more SL PRS dedicated resource pools may be (pre-)configured within an SL bandwidth part (BWP). When more than one SL PRS dedicated resource pools are (pre-)configured, more than one SL PRS dedicated resource pool may be multiplexed in a TDM and/or FDM manner or a combination thereof in the SL BWP. When more than one SL PRS dedicated resource pools are multiplexed in a TDM manner, the more than one SL PRS dedicated resource pools may be multiplexed in the same slot or different slots.

[0143] FIG. 9 illustrates a diagram **900** of multiplexing multiple SL PRS dedicated resource pools in TDM and FDM manner, in accordance with some aspects. In the example of FIG. 9, SL PRS dedicated resource pools can be multiplexed in a TDM across different slots or FDM manner.

[0144] In some aspects, the same set of parameters including the number of symbols for PSCCH and PSSCH transmission, starting symbol, number of symbols for an SL PRS transmission, comb size, and/or number of PRBs for an SL PRS transmission may be (pre-)configured as part of an SL PRS dedicated resource pool configuration. In some aspects, different sets of parameters may be (pre-)configured across different SL PRS dedicated resource pools.

[0145] In some aspects, the first subset of parameters may be the same for an SL PRS dedicated resource pool, while a second subset of parameters may be different in an SL PRS dedicated resource pool. In some aspects, the number of symbols and comb size for an SL PRS transmission may be the same for an SL PRS dedicated resource pool, while starting symbol and the number of PRBs for an SL PRS transmission may be different for different SL PRS resources in an SL PRS dedicated resource pool.

[0146] FIG. 10 illustrates a diagram **1000** of SL PRS resources with different bandwidths in an SL

PRS dedicated resource pool, in accordance with some aspects. In the example of FIG. 10, SL PRS resource #0 and resource #1 are multiplexed in an FDM manner in an SL PRS dedicated resource pool. In some aspects, different bandwidths can be (pre-)configured for different SL PRS resources in a dedicated SL PRS resource pool.

[0147] In some embodiments, when more than one SL PRS dedicated resource pools are multiplexed in an FDM manner, the number of symbols for SL PRS, the number of symbols for PSCCH/PSSCH, the comb size, the number of repetitions, and/or the number of SL PRS resources that are multiplexed in via TDM are the same across the more than one SL PRS dedicated resource pool.

[0148] In some embodiments, when more than one SL PRS dedicated resource pools are multiplexed in an FDM manner, the number of symbols and starting symbol of an SL PRS resource, the number of symbols for PSCCH/PSSCH, and the number of SL PRS resources that are multiplexed in via TDM are same across the more than one SL PRS dedicated resource pool.

[0149] In some aspects, one or more SL PRS resource pools shared with SL communications may be (pre-)configured within an SL bandwidth part (BWP). When more than one SL PRS shared resource pools are (pre-)configured, the more than one SL PRS shared resource pools may be multiplexed in a TDM and/or FDM manner or a combination thereof in the SL BWP.

[0150] In some aspects, when more than one SL PRS shared resource pools are multiplexed in a TDM manner, the more than one SL PRS shared resource pools may be multiplexed in different slots.

[0151] In some embodiments, when more than one SL PRS shared resource pools are multiplexed in an FDM manner, a number of symbols and starting symbol of the SL symbols in a slot, presence or absence of PSFCH in the slot, and a number of symbols and starting symbol of an SL PRS resource are same across the more than one SL PRS shared resource pool.

[0152] In some aspects, when more than one SL PRS shared resource pools are multiplexed in an FDM manner, the number of symbols and starting symbol of the SL symbols in a slot, presence or absence of PSFCH in the slot is the same across the more than one SL PRS shared resource pool while a number of symbols and starting symbol of an SL PRS resource may be different across the shared resource pools. In a further example, this may apply to an SL PRS shared resource pool wherein SL PRS is multiplexed with PSSCH at the RE level similar to SL CSI-RS.

[0153] In some embodiments, multiple SL PRS resources may be (pre-) configured within an SL PRS resource pool, which may depend on the number of PRBs and the number of symbols for each SL PRS resource. In some aspects, the number of PRBs, starting symbols, and the number of symbols for an SL PRS resource may be (pre-)configured by higher layers for sidelink positioning.

[0154] Further, one or more repetitions may be (pre-)configured for an SL PRS resource. In aspects when multiple repetitions of SL PRS are (pre-) configured for an SL PRS resource, the multiple repetitions may be allocated within the same SL PRS resource pool.

[0155] In some aspects, when multiple SL PRS resources or multiple repetitions of SL PRS are multiplexed in a TDM manner in an SL PRS resource pool, i.e., with a different set of symbols in a slot, the number of automatic gain control (AGC) symbols and the number of guard symbols for Tx-Rx switching purpose in an SL PRS resource may be (pre-)configured by higher layers or predetermined for an SL PRS resource pool.

[0156] In some embodiments, when more than one SL PRS resources within an SL PRS resource pool are from different UEs, AGC symbols are inserted in the first symbol of each of the multiple SL PRS resources in the SL PRS resource pool. Similarly, guard symbols are inserted in the last symbol of each SL PRS resource in the SL PRS resource pool.

[0157] FIG. 11 illustrates a diagram 1100 of SL PRS resources in an SL PRS resource pool configured according to a first option, in accordance with some aspects. In the example of FIG. 11, two SL PRS resources are configured within the SL PRS resource pool. Given that these two SL PRS resources may be transmitted from different UEs, AG symbol and guard symbol are inserted

in the first and second SL PRS resource, respectively. In some aspects, although in the examples, the starting symbol is 3rd symbol for the SL PRS resource pool, the design can be straightforwardly extended to other cases, e.g., the starting symbol for the SL PRS resource pool is the first symbol in a slot.

[0158] In some aspects, AGC symbols are inserted in the first symbol of each of the multiple SL PRS resources in the SL PRS resource pool. This may apply for the case when multiple SL PRS resources within an SL PRS resource pool are from the same UE, when different beams are applied for the more than one SL PRS resource, or when multiple SL PRS resources within an SL PRS resource pool are from the different UEs. SL PRS. In some aspects, a single guard symbol may be inserted in the last symbol of the last (in time) of the multiple SL PRS resources in the SL PRS resource pool.

[0159] FIG. 12 illustrates a diagram 1200 of SL PRS resources in an SL PRS resource pool configured according to a second option, in accordance with some aspects. In the example, two SL PRS resources are configured within the SL PRS resource pool. Given that these two SL PRS resources may be transmitted from one UE with different beam directions, a guard symbol for Tx Rx switching time can be inserted in the second SL PRS resource. In some aspects, AGC symbols are inserted in the first and second SL PRS resources, respectively.

[0160] In some aspects, when more than one SL PRS resources or more than one repetitions of an SL PRS resource within an SL PRS resource pool are from the same UE, AGC symbols are inserted in the first symbol of the more than one SL PRS resources or repetitions in the SL PRS resource pool. In addition, a single guard symbol may be inserted in the last symbol of the last (in time) of the multiple SL PRS resources or repetitions in the SL PRS resource pool.

[0161] FIG. 13 illustrates a diagram 1300 of SL PRS resources in an SL PRS resource pool configured according to a third option, in accordance with some aspects. In the example of FIG. 13, two SL PRS repetitions of an SL PRS resource are configured within the SL PRS resource pool. Given that these two SL PRS resources may be transmitted from one UE, a single guard symbol for Tx Rx switching time is inserted in the last symbol of the second SL PRS repetition. In addition, a single AGC symbol is inserted in the first symbol of the first SL PRS repetition.

SL PRS Resource Indexing Within a Resource Pool

[0162] Embodiments of SL PRS resource indexing within an SL PRS resource pool are provided as follows. The embodiments and examples in this sub-section on indexing of SL PRS resources can apply to a resource pool, regardless of whether it may be a dedicated SL PRS resource pool (with SL PRS or SL PRS and PSCCH) or a shared resource pool with SL PRS multiplexed with other SL communication channels (e.g., SL PRS, PSCCH, and PSSCH).

[0163] In some embodiments, multiple SL PRS resources may be (pre-)configured in an SL PRS resource pool. In some aspects, multiple SL PRS resources may be multiplexed in a time division multiplexing (TDM) and/or frequency division multiplexing (FDM) manner. In particular, the following multiplexing options can be used for different SL PRS resources: [0164] (a) Different comb offset indexes can be allocated for different SL PRS resources. In some aspects, for an SL PRS resource, different comb offsets may be applied for different symbols, which follows a staggered pattern similar to the DL-PRS design. In this case, an SL PRS resource may be differentiated by the starting RE offset within a comb. [0165] (b) Different frequency domain resources can be allocated for different SL PRS transmissions. For this case, a disjoint set of consecutive physical resource blocks may be allocated for different SL PRS resources. In one example, an SL PRS resource pool may span 100 PRBs, while the first 50 PRBs may be allocated for a first SL PRS resource and the second 50 PRBs may be allocated for a second SL PRS resource. [0166] (c) Different time domain resources can be allocated for different SL PRS transmissions. For this option, different SL PRS resources may be allocated in different symbols or different slots.

[0167] In some aspects, the SL PRS resource index may be determined first in an ascending order

of a starting symbol and/or a starting slot, second in an ascending order of a starting PRB, and third in an ascending order of a comb RE offset of an SL PRS transmission.

[0168] In some embodiments, if only one SL PRS in the time domain in a slot is allocated within an SL PRS resource pool, the SL PRS resource index can be determined first in ascending order of a starting PRB, and second in ascending order of a comb RE offset of an SL PRS transmission.

[0169] In some aspects, if only one SL PRS in the frequency domain in a slot is allocated within an SL PRS resource pool, the SL PRS resource index may be determined first in an ascending order of a starting symbol and/or a starting slot, and second in an ascending order of a comb RE offset of an SL PRS transmission.

[0170] In some embodiments, if only one SL PRS in the time domain and frequency in a slot is allocated within an SL PRS resource pool, the SL PRS resource index may be determined in ascending order of a comb RE offset of an SL PRS transmission.

[0171] In some embodiments, other permutations of the order for the determination of the SL PRS resource index can be straightforwardly extended from the above embodiment. In one example, the SL PRS resource index may be determined first in an ascending order of a starting PRB, second in an ascending order of a starting symbol and/or a starting slot, and third in an ascending order of a comb RE offset of an SL PRS transmission.

[0172] In some embodiments, the SL PRS resource index may be determined first in an ascending order of a comb RE offset, second in an ascending order of a starting PRB, and third in an ascending order of a starting symbol and/or a starting slot of an SL PRS transmission. In case when only one SL PRS in the frequency domain in a slot is allocated within an SL PRS resource pool, the SL PRS resource index may be determined first in an ascending order of a comb RE offset, and second in an ascending order of a starting symbol and/or a starting slot of an SL PRS transmission.

[0173] FIG. 14 illustrates a diagram 1400 of SL PRS resource determination within an SL PRS resource pool, in accordance with some aspects. In the example of FIG. 14, 12 symbols are allocated for the SL PRS resource pool, and the 1st and 7th symbols within the SL PRS resource pools are allocated for AGC symbols; and the 6th and 12th symbols within the SL PRS resource pools are allocated for guard symbol for Tx-Rx switching purpose. In some aspects, the number of AGC and guard symbols may be different for different SL PRS resource pool configurations. Further, a comb size of 2 is used for SL PRS transmission. In this example, the SL PRS resource indices are determined first in a starting symbol of an SL PRS transmission, and second, in an ascending order of an ascending order of a comb RE offset SL PRS. In this case: [0174] (a) first SL PRS resource is the SL PRS with {first symbol=2nd symbol within the SL PRS resource pool, and comb RE offset=0}; [0175] (b) second SL PRS resource is the SL PRS with {first symbol=2nd symbol within the SL PRS resource pool, and comb RE offset=1}; [0176] (c) third SL PRS resource is the SL PRS with {first symbol=8th symbol within the SL PRS resource pool, and comb RE offset=0}; and [0177] (d) fourth SL PRS resource is the SL PRS with {first symbol=8th symbol within the SL PRS resource pool, and comb RE offset=1}.

Determination of a Set of Slots for the SL PRS Resource Pool

[0178] Embodiments of the determination of a set of slots for the SL PRS resource pool are provided as follows below.

[0179] In some embodiments, a set of slots for the SL PRS resource pool can be determined after excluding the set of slots allocated for SL synchronization signal block (SSB), non-SL, reserved slots, and SL communication resource pool. In particular, the following procedure can be used to determine the set of slots for the SL PRS resource pool: the set of slots includes all the slots except the following slots: [0180] (a) Slots where the SL-SSB resource is configured. [0181] (b) Slots that are identified as non-SL slots, where some OFDM symbols are not semi-statically configured as UL in this slot as determined in Section 8 in 3GPP TS 38.214. [0182] (c) Reserved slots as determined in Section 8 in TS 38.214. [0183] (d) Slots are assigned for the SL communication resource pool as determined in Section 8 in TS 38.214.

[0184] In some aspects, a bitmap can be (pre-)configured for the SL PRS resource pool. More specifically, one or more of the following steps can be used to determine the set of slots for the SL PRS resource pool in accordance with the (pre-)configured bitmap: [0185] (a) a bitmap

$$[00001](b_0, b_1, \dots, b_{L_{\text{bitmap}}^{\text{SL-PRS}} - 1})$$

associated with the resource pool for SL PRS is used where $L_{\text{bitmap}}^{\text{SL-PRS}}$ the length of the bitmap for the SL PRS resource pool is configured by higher layers. [0186] (b) a slot $t_{\text{sub.k.sup.SL-PRS}}$

$(0 \leq k < 10240 \times 2 \times \mu - N_{\text{sub.S.sub.SSB}} - N_{\text{sub.nonSL}} - N_{\text{sub.reserved}} - N_{\text{sub.SLcomm}})$ belongs to the set if $b_{\text{sub.k}'} = 1$ where $k' = k \bmod L_{\text{bitmap}}^{\text{SL-PRS}}$, and $N_{\text{sub.SLcomm}}$ is the number of slots assigned for SL communication. [0187] (c) The slots in the set are re-indexed such that the subscripts i of the remaining slots $t'_{\text{sub.i.sup.SL-PRS}}$, are successive $\{0, 1, \dots, T'_{\text{sub.max.sup.SL-PRS}} - 1\}$ where $T'_{\text{sub.max.sup.SL-PRS}}$ is the number of slots remaining in the set.

[0188] In some embodiments, a set of slots for the SL PRS resource pool can be determined after excluding the set of slots allocated for SL SSB, non-SL, and reserved slots. In particular, the following procedure can be used to determine the set of slots for the SL PRS resource pool: the set of slots includes all the slots except the following slots: [0189] (a) Slots where the SL-SSB resource are configured; [0190] (b) Slots that are identified as non-SL slots, where some OFDM symbols are not semi-statically configured as UL in this slot as determined in Section 8 in TS 38.214; and [0191] (c) Reserved slots as determined in Section 8 in TS 38.214.

[0192] In some embodiments, a bitmap can be (pre-)configured for the SL PRS resource pool. More specifically, one or more of the following steps can be used to determine the set of slots for the SL PRS resource pool in accordance with the (pre-)configured bitmap: [0193] (a) a bitmap

$$[00002](b_0, b_1, \dots, b_{L_{\text{bitmap}}^{\text{SL-PRS}} - 1})$$

associated with the resource pool for SL PRS is used where $L_{\text{bitmap}}^{\text{SL-PRS}}$ the length of the bitmap for the SL PRS resource pool is configured by higher layers. [0194] (b) a slot $t_{\text{sub.k.sup.SL-PRS}}$ $(0 \leq k < 10240 \times 2 \times \mu - N_{\text{sub.S.sub.SSB}} - N_{\text{sub.nonSL}} - N_{\text{sub.reserved}})$ belongs to the set if $b_{\text{sub.k}'} = 1$ where $k' = k \bmod L_{\text{bitmap}}^{\text{SL-PRS}}$. [0195] (c) The slots in the set are re-indexed such that the subscripts i of the remaining slots $t'_{\text{sub.i.sup.SL-PRS}}$ are successive $\{0, 1, \dots, T'_{\text{sub.max.sup.SL-PRS}} - 1\}$ where $T'_{\text{sub.max.sup.SL-PRS}}$ is the number of slots remaining in the set.

[0196] In some embodiments, the length of the bitmap for the SL PRS resource pool may be the same as or different from that for the SL communication resource pool. When the length of the bitmap for the SL PRS resource pool is the same as that for the SL communication resource pool, disjoint sets of slots may be configured for the SL PRS resource pool and the SL communication resource pool respectively. In this case, UE may not expect that a slot is (pre-)configured for both SL PRS and SL communication resource pools. For this option, the SL PRS resource pool and the SL communication resource pool may be multiplexed in a TDM manner in different slots.

[0197] FIG. 15 illustrates diagram 1500 of the determination of a set of slots for the SL PRS resource pool according to a first option, in accordance with some aspects. In the example of FIG. 15, bitmap '0010001000' is configured for the SL PRS resource pool while bitmap '1100110000' is configured for SL communication. This indicates that $n=2, 6$ with a periodicity of 10 slots are configured for the SL PRS resource pool, and $n=0, 1, 4, 5$ with a periodicity of 10 slots are configured for the SL communication resource pool. In this case, the SL PRS resource pool and the SL communication resource pool may be multiplexed in a TDM manner in different slots.

[0198] In some embodiments, if one or more slots from the set of determined slots are the same for SL PRS and SL communication resource pool, respectively, the UE may further check whether the set of symbols and/or the set of PRBs in the one or more slots for SL PRS resource pool and SL communication overlap.

[0199] In some aspects, if the set of symbols and/or the set of PRBs in the one or more slots for the SL PRS resource pool and SL communication do not overlap, then the one or more slots can be assigned for the SL PRS resource pool. However, if the set of symbols and/or the set of PRBs in the one or more slots for the SL PRS resource pool and SL communication overlap, the one or more slots may not be assigned to the SL PRS resource pool.

[0200] FIG. 16 illustrates diagram 1600 of the determination of a set of slots for the SL PRS resource pool according to a first option, in accordance with some aspects. In the example of FIG. 16, bitmap '0100010000' is configured for the SL PRS resource pool while bitmap '1100110000' is configured for SL communication, which indicates that $n=1, 5$ with a periodicity of 10 slots are configured for the SL PRS resource pool and $n=0, 1, 4, 5$ with a periodicity of 10 slots are configured for SL communication resource pool, respectively. In some aspects, $n=1$ and 5 with a periodicity of 10 slots are configured for both SL PRS and SL communication resource pool. However, given that a disjoint set of symbols are allocated for SL PRS and SL communication resource pool, $n=1$ and 5 with a periodicity of 10 slots can be used for SL PRS. In some aspects, different SL-BWPs need to be configured for the positioning and the communication resource pool.

[0201] In some aspects, the disclosed techniques can be based on one or more of the following examples. A system and method of wireless communication for a fifth generation (5G) or new radio (NR) system include configuring, by a gNB, a resource pool for sidelink positioning reference signal (SL PRS) transmission or reception. The UE transmits an SL PRS in the resource pool for SL PRS transmission. The UE receives an SL PRS in the resource pool for SL PRS reception.

[0202] In some aspects, multiple resource pools can be configured for SL PRS transmission or reception.

[0203] In some aspects, a contiguous number of physical resource blocks (PRB) can be (pre-)configured in the frequency domain.

[0204] In some aspects, the starting PRB of an SL PRS resource pool can be (pre-)configured relative to point-A or the starting PRB of the SL bandwidth part (BWP) for SL communication or channel bandwidth.

[0205] In some aspects, the frequency domain allocation of an SL PRS resource pool may be provided to a UE via (pre-)configuration of a number of subchannels where a subchannel may be further configured with size from one of {10, 12, 15, 20, 25, 50, 75, 100} contiguous-in-frequency PRBs.

[0206] In some aspects, the starting PRB of an SL PRS resource pool can be (pre-)configured relative to the starting PRB of an SL BWP which is (pre-)configured for SL PRS transmission and reception.

[0207] In some embodiments, a starting symbol and length of symbols in a slot can be (pre-)configured in the time domain for an SL PRS resource pool.

[0208] In some aspects, subcarrier spacing and/or cyclic prefix can be (pre-)configured for an SL PRS resource pool; where subcarrier spacing and/or cyclic prefix may be the same as that (pre-)configured for SL BWP for SL communication or can be determined in accordance with the configuration from SL BWP for SL PRS transmission and reception.

[0209] In some aspects, an SL PRS resource pool may be multiplexed with an SL communication resource pool in a TDM and/or FDM manner or both.

[0210] In some aspects, an SL PRS resource pool may be (pre-)configured in a different slot from the slots that are (pre-)configured for an SL communication resource pool.

[0211] In some embodiments, an SL PRS resource pool may be (pre-)configured in different symbols in the same slot as the slots that are (pre-)configured for an SL communication resource pool.

[0212] In some aspects, multiple SL PRS resources may be configured within an SL PRS resource pool.

[0213] In some aspects, the SL PRS resource index may be determined first in an ascending order

of a starting symbol and/or a starting slot, second in an ascending order of a starting PRB, and third in an ascending order of a comb RE offset of an SL PRS transmission.

[0214] In some aspects, the SL PRS resource index can be determined first in an ascending order of a starting PRB, and second in an ascending order of a comb RE offset of an SL PRS transmission.

[0215] In some aspects, the SL PRS resource index may be determined first in an ascending order of a starting symbol and/or a starting slot, and second in an ascending order of a comb RE offset of an SL PRS transmission.

[0216] In some aspects, the SL PRS resource index may be determined in ascending order of a comb RE offset of an SL PRS transmission.

[0217] In some embodiments, the SL PRS resource index may be determined first in an ascending order of a comb RE offset, second in an ascending order of a starting PRB, and third in an ascending order of a starting symbol and/or a starting slot of an SL PRS transmission.

[0218] In some aspects, the number of automatic gain control (AGC) symbols and the number of guard symbols for Tx-Rx switching purposes in an SL PRS resource may be (pre-)configured by higher layers or predetermined for an SL PRS resource pool.

[0219] In some aspects, when more than one SL PRS resources within an SL PRS resource pool are from different UEs, AGC symbols are inserted in the first symbol of each of the multiple SL PRS resources in the SL PRS resource pool.

[0220] In some aspects, when multiple SL PRS resources within an SL PRS resource pool are from the same UE, and when different beams are applied for the more than one SL PRS resources, AGC symbols are inserted in the first symbol of each of the multiple SL PRS resources in the SL PRS resource pool.

[0221] In some aspects, when more than one SL PRS resources or more than one repetitions of an SL PRS resource within an SL PRS resource pool are from the same UE, AGC symbols are inserted in the first symbol of the more than one SL PRS resources or repetitions in the SL PRS resource pool.

[0222] In some aspects, a set of slots for the SL PRS resource pool can be determined after excluding the set of slots allocated for SL synchronization signal block (SSB), non-SL, reserved slots, and SL communication resource pool.

[0223] In some aspects, a bitmap can be (pre-)configured for the SL PRS resource pool.

[0224] In some aspects, a set of slots for the SL PRS resource pool can be determined after excluding the set of slots allocated for SL SSB, non-SL, and reserved slots.

[0225] In some aspects, the length of the bitmap for the SL PRS resource pool may be the same as or different from that for the SL communication resource pool.

[0226] In some aspects, if the set of symbols and/or the set of PRBs in the one or more slots for the SL PRS resource pool and SL communication do not overlap, then the one or more slots can be assigned for the SL PRS resource pool.

[0227] In some aspects, if the set of symbols and/or the set of PRBs in the one or more slots for the SL PRS resource pool and SL communication overlap, the one or more slots may not be assigned for the SL PRS resource pool.

[0228] In some aspects, one or more SL PRS dedicated resource pools may be (pre-)configured within an SL bandwidth part (BWP).

[0229] In some aspects, when more than one SL PRS dedicated resource pools are multiplexed in a TDM manner, the more than one SL PRS dedicated resource pools may be multiplexed in the same slot or different slots.

[0230] In some aspects, the same set of parameters including the number of symbols for PSCCH and PSSCH transmission, number of symbols for an SL PRS transmission, comb size, and/or number of PRBs for an SL PRS transmission may be (pre-)configured as part of an SL PRS dedicated resource pool configuration

[0231] In some aspects, the first subset of parameters may be the same for an SL PRS dedicated

resource pool, while a second subset of parameters may be different in an SL PRS dedicated resource pool.

[0232] In some embodiments, one or more SL PRS resource pools shared with SL communications may be (pre-)configured within an SL bandwidth part (BWP); wherein When more than one SL PRS shared resource pools are (pre-)configured, more than one SL PRS shared resource pools may be multiplexed in a TDM and/or FDM manner or a combination thereof in the SL BWP.

[0233] FIG. 17 illustrates a block diagram of a communication device such as an evolved Node-B (eNB), a new generation Node-B (gNB) (or another RAN node such as a base station), a network-controlled repeater (NCR), an access point (AP), a wireless station (STA), a mobile station (MS), or user equipment (UE), in accordance with some aspects and to perform one or more of the techniques disclosed herein. In alternative aspects, the communication device **1700** may operate as a standalone device or may be connected (e.g., networked) to other communication devices.

[0234] Circuitry (e.g., processing circuitry) is a collection of circuits implemented in tangible entities of the device **1700** that include hardware (e.g., simple circuits, gates, logic, etc.). Circuitry membership may be flexible over time. Circuitries include members that may, alone or in combination, perform specified operations when operating. In an example, the hardware of the circuitry may be immutably designed to carry out a specific operation (e.g., hardwired). In an example, the hardware of the circuitry may include variably connected physical components (e.g., execution units, transistors, simple circuits, etc.) including a machine-readable medium physically modified (e.g., magnetically, electrically, moveable placement of invariant massed particles, etc.) to encode instructions of the specific operation.

[0235] In connecting the physical components, the underlying electrical properties of a hardware constituent are changed, for example, from an insulator to a conductor or vice versa. The instructions enable embedded hardware (e.g., the execution units or a loading mechanism) to create members of the circuitry in hardware via the variable connections to carry out portions of the specific operation when in operation. Accordingly, in an example, the machine-readable medium elements are part of the circuitry or are communicatively coupled to the other components of the circuitry when the device is operating. In an example, any of the physical components may be used in more than one member of more than one circuitry. For example, under operation, execution units may be used in the first circuit of a first circuitry at one point in time and reused by a second circuit in the first circuitry, or by a third circuit in a second circuitry at a different time. Additional examples of these components with respect to the device **1700** follow.

[0236] In some aspects, the device **1700** may operate as a standalone device or may be connected (e.g., networked) to other devices. In a networked deployment, the communication device **1700** may operate in the capacity of a server communication device, a client communication device, or both in server-client network environments. In an example, the communication device **1700** may act as a peer communication device in a peer-to-peer (P2P) (or other distributed) network environment. The communication device **1700** may be a UE, eNB, PC, a tablet PC, STB, PDA, mobile telephone, smartphone, a web appliance, network router, a switch or bridge, or any communication device capable of executing instructions (sequential or otherwise) that specify actions to be taken by that communication device. Further, while only a single communication device is illustrated, the term “communication device” shall also be taken to include any collection of communication devices that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein, such as cloud computing, software as a service (SaaS), and other computer cluster configurations.

[0237] Examples, as described herein, may include, or may operate on, logic or several components, modules, or mechanisms. Modules are tangible entities (e.g., hardware) capable of performing specified operations and may be configured or arranged in a certain manner. In an example, circuits may be arranged (e.g., internally or with respect to external entities such as other circuits) in a specified manner as a module. In an example, the whole or part of one or more

computer systems (e.g., a standalone, client, or server computer system) or one or more hardware processors may be configured by firmware or software (e.g., instructions, an application portion, or an application) as a module that operates to perform specified operations. In an example, the software may reside on a communication device-readable medium. In an example, the software, when executed by the underlying hardware of the module, causes the hardware to perform the specified operations.

[0238] Accordingly, the term “module” is understood to encompass a tangible entity, be that an entity that is physically constructed, specifically configured (e.g., hardwired), or temporarily (e.g., transitorily) configured (e.g., programmed) to operate in a specified manner or to perform part or all of any operation described herein. Considering examples in which modules are temporarily configured, each of the modules needs not to be instantiated at any one moment in time. For example, where the modules comprise a general-purpose hardware processor configured using the software, the general-purpose hardware processor may be configured as respective different modules at different times. The software may accordingly configure a hardware processor, for example, to constitute a particular module at one instance of time and to constitute a different module at a different instance of time.

[0239] The communication device (e.g., UE) **1700** may include a hardware processor **1702** (e.g., a central processing unit (CPU), a graphics processing unit (GPU), a hardware processor core, or any combination thereof), a main memory **1704**, a static memory **1706**, and a storage device **1716** (e.g., hard drive, tape drive, flash storage, or other block or storage devices), some or all of which may communicate with each other via an interlink (e.g., bus) **1708**.

[0240] The communication device **1700** may further include a display device **1710**, an alphanumeric input device **1712** (e.g., a keyboard), and a user interface (UI) navigation device **1714** (e.g., a mouse). In an example, the display device **1710**, input device **1712**, and UI navigation device **1714** may be a touchscreen display. The communication device **1700** may additionally include a signal generation device **1718** (e.g., a speaker), a network interface device **1720**, and one or more sensors **1721**, such as a global positioning system (GPS) sensor, compass, accelerometer, or another sensor. The communication device **1700** may include an output controller **1728**, such as a serial (e.g., universal serial bus (USB), parallel, or other wired or wireless (e.g., infrared (IR), near field communication (NFC), etc.) connection to communicate or control one or more peripheral devices (e.g., a printer, card reader, etc.).

[0241] The storage device **1716** may include a communication device-readable medium **1722**, on which is stored one or more sets of data structures or instructions **1724** (e.g., software) embodying or utilized by any one or more of the techniques or functions described herein. In some aspects, registers of the processor **1702**, the main memory **1704**, the static memory **1706**, and/or the storage device **1716** may be, or include (completely or at least partially), the device-readable medium **1722**, on which is stored the one or more sets of data structures or instructions **1724**, embodying or utilized by any one or more of the techniques or functions described herein. In an example, one or any combination of the hardware processor **1702**, the main memory **1704**, the static memory **1706**, or the storage device **1716** may constitute the device-readable medium **1722**.

[0242] As used herein, the term “device-readable medium” is interchangeable with “computer-readable medium” or “machine-readable medium”. While the communication device-readable medium **1722** is illustrated as a single medium, the term “communication device-readable medium” may include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) configured to store the one or more instructions **1724**. The term “communication device-readable medium” is inclusive of the terms “machine-readable medium” or “computer-readable medium”, and may include any medium that is capable of storing, encoding, or carrying instructions (e.g., instructions **1724**) for execution by the communication device **1700** and that causes the communication device **1700** to perform any one or more of the techniques of the present disclosure, or that is capable of storing, encoding or carrying data structures used by or

associated with such instructions. Non-limiting communication device-readable medium examples may include solid-state memories and optical and magnetic media. Specific examples of communication device-readable media may include non-volatile memory, such as semiconductor memory devices (e.g., Electrically Programmable Read-Only Memory (EPROM), Electrically Erasable Programmable Read-Only Memory (EEPROM)) and flash memory devices; magnetic disks, such as internal hard disks and removable disks; magneto-optical disks; Random Access Memory (RAM); and CD-ROM and DVD-ROM disks. In some examples, communication device-readable media may include non-transitory communication device-readable media. In some examples, communication device-readable media may include communication device-readable media that is not a transitory propagating signal.

[0243] Instructions **1724** may further be transmitted or received over a communications network **1726** using a transmission medium via the network interface device **1720** utilizing any one of several transfer protocols. In an example, the network interface device **1720** may include one or more physical jacks (e.g., Ethernet, coaxial, or phone jacks) or one or more antennas to connect to the communications network **1726**. In an example, the network interface device **1720** may include a plurality of antennas to wirelessly communicate using at least one of the single-input-multiple-output (SIMO), MIMO, or multiple-input-single-output (MISO) techniques. In some examples, the network interface device **1720** may wirelessly communicate using Multiple User MIMO techniques.

[0244] The term “transmission medium” shall be taken to include any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the communication device **1700**, and includes digital or analog communications signals or another intangible medium to facilitate communication of such software. In this regard, a transmission medium in the context of this disclosure is a device-readable medium.

[0245] The terms “machine-readable medium,” “computer-readable medium,” and “device-readable medium” mean the same thing and may be used interchangeably in this disclosure. The terms are defined to include both machine-storage media and transmission media. Thus, the terms include both storage devices/media and carrier waves/modulated data signals.

[0246] Described implementations of the subject matter can include one or more features, alone or in combination as illustrated below by way of examples.

[0247] Example 1 is an apparatus for user equipment (UE) configured for operation in a Fifth Generation New Radio (5G NR) network, the apparatus comprising: processing circuitry, wherein to configure the UE for sidelink positioning in the 5G NR network, the processing circuitry is to: decode at least one of radio resource control (RRC) signaling, LTE positioning protocol (LPP) signaling received from a base station for in-coverage scenario, sidelink positioning protocol (SLPP) signaling received from a second UE, or pre-configuration for out-of-coverage scenario, the LPP signaling, the RRC signaling, the SLPP signaling or the pre-configuration respectively including configuration information configuring a first resource pool associated with sidelink communications and a second resource pool associated with sidelink positioning reference signal, the first resource pool being multiplexed with the second resource pool; encode sidelink data for a sidelink transmission to a second UE using a first time-frequency resource from the first resource pool; and encode a first sidelink positioning reference signal (SL PRS) for transmission to a third UE using a second time-frequency resource from the second resource pool; and a memory coupled to the processing circuitry and configured to store the configuration information.

[0248] In Example 2, the subject matter of Example 1 includes subject matter where the processing circuitry is to: decode the LPP signaling, the RRC signaling, the SLPP signaling, or the pre-configuration to obtain multiple resource pools for the first SL PRS, the multiple resource pools including the second resource pool and at least a third resource pool, for transmission and/or reception of SL PRSs.

[0249] In Example 3, the subject matter of Example 2 includes subject matter where the processing

circuitry is to: decode a positioning measurement received from the third UE using a third time-frequency resource from the third resource pool, the positioning measurement corresponding to the first SL PRS.

[0250] In Example 4, the subject matter of Examples 2-3 includes subject matter where the processing circuitry is to: decode a second SL PRS received from the first UE using a third time-frequency resource from the third resource pool; and encode a positioning measurement for transmission to a Location Management Function (LMF) entity or the second UE, the positioning measurement corresponding to the second SL PRS.

[0251] In Example 5, the subject matter of Examples 2-4 includes subject matter where the multiple resource pools are multiplexed in one or both of a time-division multiplexing (TDM) manner and a frequency-division multiplexing (FDM) manner.

[0252] In Example 6, the subject matter of Example 5 includes subject matter where the multiple resource pools are multiplexed in the TDM manner, and each resource pool of the multiple resource pools occupies a separate slot.

[0253] In Example 7, the subject matter of Examples 1-6 includes subject matter where at least the second resource pool is an SL PRS dedicated resource pool that is configured within a sidelink (SL) bandwidth part (BWP).

[0254] In Example 8, the subject matter of Examples 1-7 includes subject matter where the configuration information configures a plurality of SL PRS dedicated resource pools that are multiplexed in a time-division multiplexing (TDM) manner.

[0255] In Example 9, the subject matter of Example 8 includes subject matter where at least two SL PRS dedicated resource pools of the plurality of SL PRS dedicated resource pools are multiplexed in a same slot.

[0256] In Example 10, the subject matter of Examples 1-9 includes subject matter where the configuration information includes an SL PRS dedicated resource pool configuration including one or more of: a number of symbols for physical sidelink control channel (PSCCH) a starting symbol, number of symbols for an SL PRS transmission, comb size for the SL PRS transmission, and/or a number of physical resource blocks (PRBs) for the SL PRS transmission.

[0257] Example 11 is a computer-readable storage medium that stores instructions for execution by one or more processors of a base station, the instructions to configure the base station for sidelink positioning communication in a Fifth Generation New Radio (5G NR) network, and to cause the base station to perform operations comprising: encoding radio resource control (RRC) signaling for transmission to at least one user equipment (UE), the RRC signaling including configuration information configuring a first resource pool associated with sidelink communications and a second resource pool associated with sidelink positioning reference signal (SL PRS), the first resource pool being multiplexed with the second resource pool.

[0258] In Example 12, the subject matter of Example 11 includes, the operations further comprising: encoding the RRC signaling to include multiple resource pools for the sidelink positioning reference signal (SL PRS), the multiple resource pools including the second resource pool, and at least a third resource pool, for transmission and/or reception of SL PRSs, wherein the multiple resource pools are multiplexed in one or both of a time-division multiplexing (TDM) manner and a frequency-division multiplexing (FDM) manner.

[0259] Example 13 is a computer-readable storage medium that stores instructions for execution by one or more processors of user equipment (UE), the instructions to configure the UE for sidelink positioning communication in a Fifth Generation New Radio (5G NR) network, and to cause the UE to perform operations comprising: decoding radio resource control (RRC) signaling or LTE positioning protocol (LPP) signaling received from a base station for in-coverage scenario, or sidelink positioning protocol (SLPP) signaling from a second UE, or pre-configuration for out-of-coverage scenario, the LPP, RRC or SLPP signaling or the pre-configuration respectively including configuration information configuring a first resource pool associated with sidelink

communications and a second resource pool associated with sidelink positioning reference signal (SL PRS), the first resource pool being multiplexed with the second resource pool; encoding sidelink data for a sidelink transmission to a second UE using a first time-frequency resource from the first resource pool; and encoding a first sidelink positioning reference signal (SL PRS) for transmission to a third UE using a second time-frequency resource from the second resource pool. [0260] In Example 14, the subject matter of Example 13 includes, the operations further comprising: decoding the LPP signaling, the RRC signaling, the SLPP signaling, or the pre-configuration to obtain multiple resource pools for the sidelink positioning communication, the multiple resource pools including the second resource pool and at least a third resource pool, for transmission and/or reception of SL PRSs.

[0261] In Example 15, the subject matter of Example 14 includes, the operations further comprising: decoding a positioning measurement received from the third UE using a third time-frequency resource from the third resource pool, the positioning measurement corresponding to the first SL PRS.

[0262] In Example 16, the subject matter of Examples 14-15 includes, the operations further comprising: decoding a second SL PRS received from the third UE using a third time-frequency resource from the third resource pool; and encoding a positioning measurement for transmission to a Location Management Function (LMF) entity or the second UE, the positioning measurement corresponding to the second SL PRS.

[0263] In Example 17, the subject matter of Examples 14-16 includes subject matter where the multiple resource pools are multiplexed in one or both of a time-division multiplexing (TDM) manner and a frequency-division multiplexing (FDM) manner.

[0264] In Example 18, the subject matter of Example 17 includes subject matter where the multiple resource pools are multiplexed in the TDM manner and each resource pool of the multiple resource pools occupies a separate slot.

[0265] In Example 19, the subject matter of Examples 13-18 includes subject matter where at least the second resource pool is an SL PRS dedicated resource pool that is configured within a sidelink (SL) bandwidth part (BWP).

[0266] In Example 20, the subject matter of Examples 13-19 includes subject matter where the configuration information configures a plurality of SL PRS dedicated resource pools that are multiplexed in a time-division multiplexing (TDM) manner, and wherein at least two SL PRS dedicated resource pools of the plurality of SL PRS dedicated resource pools are multiplexed in a same slot.

[0267] Example 21 is at least one machine-readable medium including instructions that, when executed by processing circuitry, cause the processing circuitry to perform operations to implement any of Examples 1-20.

[0268] Example 22 is an apparatus comprising means to implement any of Examples 1-20.

[0269] Example 23 is a system to implement any of Examples 1-20.

[0270] Example 24 is a method to implement any of Examples 1-20.

[0271] Although an aspect has been described concerning specific exemplary aspects, it will be evident that various modifications and changes may be made to these aspects without departing from the broader scope of the present disclosure. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense. This Detailed Description, therefore, is not to be taken in a limiting sense, and the scope of various aspects is defined only by the appended claims, along with the full range of equivalents to which such claims are entitled.

Claims

1-20. (canceled)

21. An apparatus for a user equipment (UE) configured for operation in a Fifth Generation New

Radio (5G NR) network, the apparatus comprising: processing circuitry, wherein to configure the UE for sidelink positioning in the 5G NR network, the processing circuitry is to: decode higher layer signaling received from a base station, the higher layer signaling to configure a plurality of sidelink positioning reference signal (SL PRS) resource pools; select an SL PRS resource from an SL PRS resource pool of the plurality of SL PRS resource pools; decode configuration signaling to determine a comb offset of the SL PRS resource; and encode an SL PRS for transmission to a second UE based on the SL PRS resource and the comb offset; and a memory coupled to the processing circuitry and configured to store the higher layer signaling.

22. The apparatus of claim 21, wherein the configuration signaling indicates one or more additional comb offsets for one or more remaining SL PRS resources of the SL PRS resource pool.

23. The apparatus of claim 21, wherein the SL PRS resource pool is a dedicated SL PRS resource pool.

24. The apparatus of claim 21, wherein the processing circuitry is to: decode at least one of radio resource control (RRC) signaling, LTE positioning protocol (LPP) signaling received from the base station for an in-coverage scenario, sidelink positioning protocol (SLPP) signaling received from the second UE, or pre-configuration for an out-of-coverage scenario, the LPP signaling, the RRC signaling, the SLPP signaling or the pre-configuration respectively including configuration information configuring a first additional resource pool associated with sidelink communications and a second additional resource pool associated with the SL PRS, the first additional resource pool being multiplexed with the second additional resource pool; encode sidelink data for a sidelink transmission to the second UE using a first time-frequency resource from the first additional resource pool; and encode a second SL PRS for transmission to a third UE using a second time-frequency resource from the second additional resource pool.

25. The apparatus of claim 24, wherein the processing circuitry is to: decode the LPP signaling, the RRC signaling, the SLPP signaling, or the pre-configuration to obtain multiple resource pools for the SL PRS, the multiple resource pools including the second additional resource pool and at least a third additional resource pool, for transmission and/or reception of SL PRSs.

26. The apparatus of claim 25, wherein the processing circuitry is to: decode a positioning measurement received from the third UE using a third time-frequency resource from the third additional resource pool, the positioning measurement corresponding to the SL PRS.

27. The apparatus of claim 25, wherein the processing circuitry is to: decode a second SL PRS received from the first UE using a third time-frequency resource from the third additional resource pool; and encode a positioning measurement for transmission to a Location Management Function (LMF) entity or the second UE, the positioning measurement corresponding to the second SL PRS.

28. The apparatus of claim 25, wherein the multiple resource pools are multiplexed in one or both of a time-division multiplexing (TDM) manner and a frequency-division multiplexing (FDM) manner.

29. The apparatus of claim 28, wherein the multiple resource pools are multiplexed in the TDM manner, and each resource pool of the multiple resource pools occupies a separate slot.

30. The apparatus of claim 21, further comprising: transceiver circuitry coupled to the processing circuitry; and one or more antennas coupled to the transceiver circuitry.

31. A non-transitory computer-readable storage medium that stores instructions for execution by one or more processors of a user equipment (UE), the instructions to configure the UE for sidelink positioning in a Fifth Generation New Radio (5G NR) network, and to cause the UE to perform operations comprising: decoding higher layer signaling received from a base station, the higher layer signaling to configure a plurality of sidelink positioning reference signal (SL PRS) resource pools; selecting an SL PRS resource from an SL PRS resource pool of the plurality of SL PRS resource pools; decoding configuration signaling to determine a comb offset of the SL PRS resource; and encoding an SL PRS for transmission to a second UE based on the SL PRS resource and the comb offset.

- 32.** The non-transitory computer-readable storage medium of claim 31, wherein the configuration signaling indicates one or more additional comb offsets for one or more remaining SL PRS resources of the SL PRS resource pool.
- 33.** The non-transitory computer-readable storage medium of claim 31, wherein the SL PRS resource pool is a dedicated SL PRS resource pool.
- 34.** The non-transitory computer-readable storage medium of claim 31, the operations comprising: decoding at least one of radio resource control (RRC) signaling, LTE positioning protocol (LPP) signaling received from the base station for an in-coverage scenario, sidelink positioning protocol (SLPP) signaling received from the second UE, or pre-configuration for an out-of-coverage scenario, the LPP signaling, the RRC signaling, the SLPP signaling or the pre-configuration respectively including configuration information configuring a first additional resource pool associated with sidelink communications and a second additional resource pool associated with the SL PRS, the first additional resource pool being multiplexed with the second additional resource pool; encoding sidelink data for a sidelink transmission to the second UE using a first time-frequency resource from the first additional resource pool; and encoding a second SL PRS for transmission to a third UE using a second time-frequency resource from the second additional resource pool.
- 35.** The non-transitory computer-readable storage medium of claim 34, the operations comprising: decoding the LPP signaling, the RRC signaling, the SLPP signaling, or the pre-configuration to obtain multiple resource pools for the SL PRS, the multiple resource pools including the second additional resource pool and at least a third additional resource pool, for transmission and/or reception of SL PRSs.
- 36.** The non-transitory computer-readable storage medium of claim 35, the operations comprising: decoding a positioning measurement received from the third UE using a third time-frequency resource from the third additional resource pool, the positioning measurement corresponding to the SL PRS.
- 37.** The non-transitory computer-readable storage medium of claim 35, the operations comprising: decoding a second SL PRS received from the first UE using a third time-frequency resource from the third additional resource pool; and encoding a positioning measurement for transmission to a Location Management Function (LMF) entity or the second UE, the positioning measurement corresponding to the second SL PRS.
- 38.** A user equipment (UE) configured for operation in a Fifth Generation New Radio (5G NR) network, the UE comprising: front-end circuitry coupled to one or more antennas; processing circuitry coupled to the front-end circuitry, the processing circuitry is to: decode higher layer signaling received from a base station, the higher layer signaling to configure a plurality of sidelink positioning reference signal (SL PRS) resource pools; select an SL PRS resource from an SL PRS resource pool of the plurality of SL PRS resource pools; decode configuration signaling to determine a comb offset of the SL PRS resource; and encode an SL PRS for transmission to a second UE based on the SL PRS resource and the comb offset; and a memory coupled to the processing circuitry and configured to store the higher layer signaling.
- 39.** The UE of claim 38, wherein the configuration signaling indicates one or more additional comb offsets for one or more remaining SL PRS resources of the SL PRS resource pool.
- 40.** The UE of claim 38, wherein the SL PRS resource pool is a dedicated SL PRS resource pool.
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